

Complex Analysis

Qualifying Exam Study Guide

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Part I

Practice Questions

How to Use the Practice List

This list is based on all 166 exercises in the repository:

- 9 questions in `Exams/main.tex`;
- 50 questions from the August 2018, August 2020, August 2022, August 2024, and January 2025 qualifying exams;
- 107 textbook exercises in `Exercise/complex.tex`.

The ordering gives the most weight to questions that appeared on a qualifying exam, especially when essentially the same textbook exercise later appeared on an exam. Check an item only after you can solve it from a blank page, justify every theorem's hypotheses, and explain the proof aloud.

Essential / Top 12

The first results and proof techniques to master.

Important advanced

Material needed for broad Part II coverage.

Warm-up

Shorter exercises for fluency and review.

Essential Shortlist

1. Prove the Fundamental Theorem of Algebra by completing the following two steps for a nonconstant polynomial

$$p(z) = a_0 + a_1z + \cdots + a_nz^n,$$

where $a_n \neq 0$ and $n \geq 1$.

- (a) Show that there is a nonzero polynomial $q(z)$ such that

$$\frac{1}{zp(z)} = \frac{1}{zp(0)} + \frac{q(z)}{p(z)},$$

for $z \neq 0$ and $p(z) \neq 0$.

- (b) By means of integration over the circle $C_R = Re^{it}$ with $t \in [0, 2\pi]$, show that $p(z_0) = 0$ for some point z_0 in $\mathbb{C}$. *Source: January 2025, I.1*

2. How many roots does the polynomial

$$z^9 + z^5 - 8z^3 + 2z + 1$$

have between the circles $\{|z| = 1\}$ and $\{|z| = 2\}$?

Source: August 2024, I.3

3. Show that the equation

$$e^z = cz,$$

where c is a constant with $|c| > e$, has exactly one solution in the set

$$\{z \in \mathbb{C} : \operatorname{Re} z < 1\}.$$

Source: January 2025, I.2

4. Suppose f is a holomorphic function on the open unit disc

$$D = \{z : |z| < 1\}.$$

Prove that there is a sequence $(z_n)_{n=1}^\infty$ in D such that $|z_n| \rightarrow 1$ and the set

$$\{f(z_n) : n \geq 1\}$$

is bounded.

Source: August 2018, I.3

5. Show that if $f : \mathbb{C} \rightarrow \mathbb{C}$ is a continuous function such that f is analytic off $[-1, 1]$, then f is an entire function.

Source: January 2025, I.3

6. Suppose that F and G are holomorphic functions on the right half plane

$$H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}.$$

Suppose also that F and G have the same zeros with the same multiplicities at each zero. Prove that there is a holomorphic function h on H such that

$$F(z) = e^{h(z)}G(z)$$

for all $z \in H$.

Source: August 2018, I.2

7. Find a conformal map f from the vertical strip

$$\Omega = \{z \in \mathbb{C} : 3 < \operatorname{Re}(z) < 11\}$$

onto the open unit disk $\mathbb{D}$.

Source: August 2018, I.1

8. Let

$$G = \{z : \operatorname{Re}(z) > 0\},$$

and $f : G \rightarrow G$ be a one-one analytic function from G into G . Suppose $f(a) = a$ for some positive real number a . Show that

$$|f'(a)| \leq 1.$$

Source: August 2020, II.4

9. Let $\mathfrak{F}$ be the collection of holomorphic functions on the open unit disk

$$D = \{z : |z| < 1\}$$

satisfying

$$\iint_D |f| \, dx \, dy \leq 1.$$

Prove or disprove that $\mathfrak{F}$ is a normal family.

Source: August 2018, II.5

10. Suppose that G is a region in $\mathbb{C}$, $\{f_n\}$ is a sequence in $H(G)$, f is a nonconstant function, and $f_n \rightarrow f$ in $H(G)$. Let $a \in G$ and $b = f(a)$. Show that there is a sequence $\{a_n\}$ in G such that

(i) $a = \lim a_n$,

(ii) $f_n(a_n) = b$ for sufficiently large n . *Source: August 2020, II.3*

11. Given entire functions f, g without common zeros, prove that there exist entire functions A, B such that

$$Af + Bg = 1.$$

Hint: Use the Mittag-Leffler Theorem.

Source: August 2020, II.1

12. Let G be a domain in the complex plane and $u : G \rightarrow \mathbb{R}$ be a nonconstant harmonic function and let

$$A = \{z \in G : u_x(z) = u_y(z) = 0\}.$$

Show that A does not have any limit points in G .

Source: January 2025, II.2

Core Qualifying-Exam Topics

3.1 Fundamental Theorem of Algebra by several methods

Exercise 3.1.1. Using the maximum modulus principle, prove the Fundamental Theorem of Algebra.

Source: August 2020, I.4

Exercise 3.1.2. Using Liouville's Theorem, prove the Fundamental Theorem of Algebra.

Source: January 2025, I.4

Exercise 3.1.3. Prove the Fundamental Theorem of Algebra by completing the following two steps for a nonconstant polynomial

$$p(z) = a_0 + a_1z + \cdots + a_nz^n,$$

where $a_n \neq 0$ and $n \geq 1$.

(a) Show that there is a nonzero polynomial $q(z)$ such that

$$\frac{1}{zp(z)} = \frac{1}{zp(0)} + \frac{q(z)}{p(z)},$$

for $z \neq 0$ and $p(z) \neq 0$.

(b) By means of integration over the circle $C_R = Re^{it}$ with $t \in [0, 2\pi]$, show that $p(z_0) = 0$ for some point z_0 in $\mathbb{C}$. *Source: January 2025, I.1*

Why: the FTA appeared three times in five recorded qualifying exams, with three different requested proofs.

3.2 Rouché's theorem and counting zeros

Exercise 3.2.1. How many roots does the polynomial

$$z^9 + z^5 - 8z^3 + 2z + 1$$

have between the circles $\{|z| = 1\}$ and $\{|z| = 2\}$?

Source: August 2024, I.3

Exercise 3.2.2. (a) Let $a > 1$. Show that the equation

$$ze^{a-z} = 1$$

has precisely one root in the closed unit disk $\{z : |z| \leq 1\}$.

(b) Show that the root in part (a) is positive. *Source: August 2020, I.3*

Exercise 3.2.3. Show that the equation

$$e^z = cz,$$

where c is a constant with $|c| > e$, has exactly one solution in the set

$$\{z \in \mathbb{C} : \operatorname{Re} z < 1\}.$$

Source: January 2025, I.2

Exercise 3.2.4. Suppose f is analytic on $\overline{B(0;1)}$ and satisfies $|f(z)| < 1$ for $|z| = 1$. Find the number of solutions (counting multiplicities) of the equation

$$f(z) = z^n$$

where n is an integer larger than or equal to 1.

Source: text exercise

Exercise 3.2.5. Determine the number of solutions, counted with multiplicity, of

$$z^6 + 7z^3 - 2z + 3 = 0$$

that satisfy $1 < |z| < 2$.

Source: August 2022, II.1

Why: you should be able to choose the comparison function and contour yourself, not only recognize Rouché after being told to use it.

3.3 Liouville, growth, and maximum/minimum modulus arguments

Exercise 3.3.1. Let f be an entire function. Show that f is a constant function if there exist $M > 0$ and $R > 0$ such that

$$|f(z)| \leq M|z|^{1/2} \quad \text{whenever } |z| > R.$$

Source: August 2020, I.1

Exercise 3.3.2. Let f be an entire function and suppose there is a constant M , an $R > 0$, and an integer $n \geq 1$ such that

$$|f(z)| \leq M|z|^n \quad \text{for } |z| > R.$$

Show that f is a polynomial of degree $\leq n$.

Source: text exercise

Exercise 3.3.3. Let f and g be entire functions satisfying $f(0) = g(0) \neq 0$ and

$$|f(z)| \leq |g(z)| \quad \forall z \in \mathbb{C}.$$

Show that $f = g$.

Source: August 2018, I.5

Exercise 3.3.4. Let G be a region and suppose that $f : G \rightarrow \mathbb{C}$ is analytic and $a \in G$ such that

$$|f(a)| \leq |f(z)| \quad \text{for all } z \in G.$$

Show that either $f(a) = 0$ or f is constant.

Source: text exercise

3.4 Boundary blow-up contradiction on the disk

Exercise 3.4.1. Suppose f is a holomorphic function on the open unit disc

$$D = \{z : |z| < 1\}.$$

Prove that there is a sequence $(z_n)_{n=1}^\infty$ in D such that $|z_n| \rightarrow 1$ and the set

$$\{f(z_n) : n \geq 1\}$$

is bounded.

Source: August 2018, I.3

Exercise 3.4.2. Let D be the open unit disk. Show that no analytic function $f : D \rightarrow \mathbb{C}$ has the property that

$$|f(z_n)| \rightarrow \infty, \quad \text{whenever } |z_n| \rightarrow 1.$$

Source: August 2024, I.2

Why: this exact idea returned six years later. Know how to remove the finitely many zeros and apply the maximum modulus principle to the reciprocal.

3.5 Cauchy's formula, residues, and winding numbers

Exercise 3.5.1. Define $\gamma : [0, 2\pi] \rightarrow \mathbb{C}$ by

$$\gamma(t) = e^{it}.$$

Compute

$$\int_{\gamma} z^n dz$$

for every integer n .

Source: Fall 2025, I.1–2

Exercise 3.5.2. Compute the integral

$$\int_{\gamma} \frac{e^z}{(z-2)^2(z-5)^5} dz,$$

where γ is the closed path

$$\gamma(t) = 3e^{it}, \quad 0 \leq t \leq 2\pi.$$

Source: Fall 2025, II.1–2

Exercise 3.5.3. Fix $w = re^{i\theta} \neq 0$ and let γ be a piecewise smooth curve in $\mathbb{C} \setminus \{0\}$ from 1 to w . Show that there is an integer k such that

$$\int_{\gamma} \frac{1}{z} dz = \log r + i\theta + 2\pi ik.$$

Source: August 2020, I.2

Exercise 3.5.4. Let G be a region and $z_0 \in G$. Suppose $f : G \rightarrow \mathbb{C}$ is analytic and $f'(z_0) \neq 0$. Show that

$$\int_{\gamma} \frac{1}{f(z) - f(z_0)} dz = \frac{2\pi i}{f'(z_0)}$$

if $\gamma = \gamma(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$ and $r > 0$ is small enough.

Source: January 2025, I.5

Exercise 3.5.5. Evaluate the integral $\int_{\gamma} \frac{dz}{z^2 + 1}$ where $\gamma(\theta) = 2|\cos 2\theta|e^{i\theta}$ for $0 \leq \theta \leq 2\pi$.

Source: text exercises

Exercise 3.5.6. Let $a > 0$. Compute

$$\int_{-\infty}^{\infty} \frac{\sin(ax)}{x(x^2 + 1)} dx$$

by using residue theory.

Source: August 2022, I.3

3.6 Removable sets and Morera's theorem

Exercise 3.6.1. Show that if $f : \mathbb{C} \rightarrow \mathbb{C}$ is a continuous function such that f is analytic off $[-1, 1]$, then f is an entire function.

Source: January 2025, I.3

Why: be ready to subdivide triangles intersecting the segment and use continuity to control the thin pieces.

3.7 Analytic logarithms, zeros, and branches

Exercise 3.7.1. Suppose that F and G are holomorphic functions on the right half plane

$$H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}.$$

Suppose also that F and G have the same zeros with the same multiplicities at each zero. Prove that there is a holomorphic function h on H such that

$$F(z) = e^{h(z)}G(z)$$

for all $z \in H$.

Source: August 2018, I.2

Exercise 3.7.2. Let $0 < R_1 < R_2$ and $\alpha \in \mathbb{R}$, and let G be the annulus centered at the origin with radii R_1 and R_2 . Let

$$z^\alpha = \exp(\alpha \log z),$$

where $\log z$ is a branch of the logarithm. Prove that z^α is analytic in G if and only if α is an integer.

Source: August 2020, II.5

Exercise 3.7.3. Give the principal branch of

$$\sqrt{1 - z^3}.$$

Source: Fall 2025, I.7

Exercise 3.7.4. Prove that there is no branch of the logarithm defined on $G = \mathbb{C} \setminus \{0\}$. (*Hint: Suppose such a branch exists and compare this with the principal branch.*)

Source: text exercise

3.8 Conformal maps and Möbius transformations

Exercise 3.8.1. Find a conformal map f from the vertical strip

$$\Omega = \{z \in \mathbb{C} : 3 < \operatorname{Re}(z) < 11\}$$

onto the open unit disk $\mathbb{D}$.

Source: August 2018, I.1

Exercise 3.8.2. Map

$$G = \mathbb{C} \setminus \{z : -9 \leq z \leq 9\}$$

onto the open unit disk by an analytic function f . Can f be one-to-one?

Source: Fall 2025, I.6

Exercise 3.8.3. Let $D = \{z : |z| < 1\}$ and find all Möbius transformations T such that $T(D) = D$.

Source: text exercise

Exercise 3.8.4. (1) Show that a Möbius transformation has 0 and ∞ as its only fixed points if and only if it is a dilation, but not the identity.

(2) Show that a Möbius transformation has ∞ as its only fixed point if and only if it is a translation, but not the identity. *Source: Fall 2025, I.3*

Exercise 3.8.5. Can you map the open unit disk conformally onto $\{z : 0 < |z| < 1\}$?

Source: text exercise

Exercise 3.8.6. Find a conformal map from the vertical strip

$$\{z \in \mathbb{C} : -2 < \operatorname{Re}(z) < 2\}$$

onto the open unit disk $D(0, 1)$. Give an explicit formula for your map.

Source: August 2022, I.2

3.9 Schwarz lemma and half-plane self-maps

Exercise 3.9.1. Let

$$G = \{z : \operatorname{Re}(z) > 0\},$$

and $f : G \rightarrow G$ be a one-one analytic function from G into G . Suppose $f(a) = a$ for some positive real number a . Show that

$$|f'(a)| \leq 1.$$

Source: August 2020, II.4

Exercise 3.9.2. Suppose $|f(z)| \leq 1$ for $|z| < 1$ and f is a non-constant analytic function. By considering the function $g : \mathbb{D} \rightarrow \mathbb{D}$ defined by

$$g(z) = \frac{f(z) - a}{1 - \overline{a}f(z)}$$

where $a = f(0)$, prove that

$$\frac{|f(0)| - |z|}{1 + |f(0)||z|} \leq |f(z)| \leq \frac{|f(0)| + |z|}{1 - |f(0)||z|}$$

for $|z| < 1$.

Source: text exercise

Exercise 3.9.3. Suppose $f : \mathbb{D} \rightarrow \mathbb{C}$ satisfies $\Re f(z) \geq 0$ for all $z \in \mathbb{D}$ and suppose that f is analytic and not constant.

(1) Show that $\Re f(z) > 0$ for all $z \in \mathbb{D}$.

(2) By using an appropriate Möbius transformation, apply Schwarz's Lemma to prove that if $f(0) = 1$ then

$$|f(z)| \leq \frac{1 + |z|}{1 - |z|}$$

for $|z| < 1$. What can be said if $f(0) \neq 1$?

(3) Show that if $f(0) = 1$, f also satisfies

$$|f(z)| \geq \frac{1 - |z|}{1 + |z|}.$$

Hint: Use part (a). *Source: text exercise*

Exercise 3.9.4. Let f be analytic on the open square

$$\Omega = \{z = x + iy : x, y \in (-1, 1)\}$$

and suppose that $|f| \leq 1$ on Ω . Show that

$$|f'(0)| < 1.$$

Source: August 2022, I.1

3.10 Normal families from estimates

Exercise 3.10.1. Let $\mathfrak{F}$ be the collection of holomorphic functions on the open unit disk

$$D = \{z : |z| < 1\}$$

satisfying

$$\iint_D |f| \, dx \, dy \leq 1.$$

Prove or disprove that $\mathfrak{F}$ is a normal family.

Source: August 2018, II.5

Why: this requires converting an area integral into a uniform bound on compact subsets, typically through the mean-value property or Cauchy estimates.

3.11 Operations on normal families

Exercise 3.11.1. Show that if $\mathcal{F} \subset H(G)$ is normal then

$$\mathcal{F}' = \{f' : f \in \mathcal{F}\}$$

is also normal.

Source: January 2025, II.5

Exercise 3.11.2. Let G be a region and suppose $\mathcal{F}$ is normal in $H(G)$ and Ω is open in $\mathbb{C}$ such that $f(G) \subset \Omega$ for every $f \in \mathcal{F}$. Show that if g is analytic on Ω and is bounded on bounded sets then

$$\{g \circ f : f \in \mathcal{F}\}$$

is normal.

Source: August 2024, II.3

Exercise 3.11.3. Let G be a region and $f, f_1, f_2, \dots$ be elements of $H(G)$. Show that $f_n \rightarrow f$ in $H(G)$ if and only if for each closed rectifiable curve γ in G , $f_n \rightarrow f$ uniformly on γ .

Source: August 2024, II.2

3.12 Hurwitz-type convergence and moving points

Exercise 3.12.1. Suppose that G is a region in $\mathbb{C}$, $\{f_n\}$ is a sequence in $H(G)$, f is a nonconstant function, and $f_n \rightarrow f$ in $H(G)$. Let $a \in G$ and $b = f(a)$. Show that there is a sequence $\{a_n\}$ in G such that

(i) $a = \lim a_n$,

(ii) $f_n(a_n) = b$ for sufficiently large n . *Source: August 2020, II.3*

Exercise 3.12.2. Let $\{f_n\} \subset H(G)$ be a sequence of one-one functions which converge to f . If G is a region, show that either f is one-one or f is a constant function.

Source: text exercise

Exercise 3.12.3. Let G and Ω be two domains in the complex plane. Suppose $\{f_n\}$ is a sequence in $C(G, \Omega)$ which converges to f and $\{z_n\}$ is a sequence in G which converges to a point z in G . Show

$$\lim_{n \rightarrow \infty} f_n(z_n) = f(z).$$

Source: January 2025, II.4

Exercise 3.12.4. Let (f_n) be holomorphic on a domain Ω , converge locally uniformly to f , and satisfy $f_n(\Omega) \subseteq \Omega$. Show that either $f(\Omega) \subseteq \Omega$, or $f(\Omega)$ consists of a single point in $\partial\Omega$.

Source: August 2022, I.5

Exercise 3.12.5. Show that, given any $\varepsilon > 0$, for sufficiently large n , all the zeros of

$$f_n(z) = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \cdots + \frac{1}{n!z^n}$$

lie inside $D(0, \varepsilon)$.

Source: August 2022, II.5

3.13 Weierstrass products and prescribed zeros

Exercise 3.13.1. Find an entire function f such that for every integer $n \geq 1$, f has n as a zero of order n . Find the most elementary solution possible.

Source: August 2024, II.5

Exercise 3.13.2. Find an entire function f such that

$$f\left(ne^{2k\pi i/n}\right) = 0$$

for all positive integers n and all integers $0 \leq k \leq n-1$. Give the most elementary example possible.

Source: January 2025, II.3

Exercise 3.13.3. Show there is an analytic function f on

$$D = \{z : |z| < 1\}$$

which is not analytic on any region G which properly contains D .

Source: August 2024, II.4

Exercise 3.13.4. Find an entire function f such that $f(m+in) = 0$ for all possible integers m, n . Find the most elementary solution possible.

Source: text exercise

Exercise 3.13.5. Construct an entire function f satisfying:

- (a) f has simple zeros at each $\sqrt{n}$, $n = 1, 2, \dots$;
- (b) f has double zeros at each $\pm i\sqrt{k}$, $k = 0, 1, 2, \dots$;
- (c) f has no other zeros. *Source: August 2022, II.2*

3.14 Mittag-Leffler and analytic Bézout identities

Exercise 3.14.1. Given entire functions f, g without common zeros, prove that there exist entire functions A, B such that

$$Af + Bg = 1.$$

Hint: Use the Mittag-Leffler Theorem.

Source: August 2020, II.1

Exercise 3.14.2. Let G be a region and let $\{a_n\}$ and $\{b_m\}$ be two sequences of distinct points in G without limit points in G such that $a_n \neq b_m$ for all n, m . Let $S_n(z)$ be a singular part at a_n and let p_m be a positive integer. Show that there is a meromorphic function f on G whose poles and zeros are $\{a_n\}$ and $\{b_m\}$ respectively, the singular part at $z = a_n$ is $S_n(z)$, and $z = b_m$ is a zero of multiplicity p_m .

Source: text exercise

3.15 Basic analyticity, inverse functions, and reflection

Exercise 3.15.1. Let G and Ω be two regions in $\mathbb{C}$, and $h : G \rightarrow \mathbb{C}$ be a one-one analytic function, and $f : G \rightarrow \mathbb{C}$ be a continuous function with $f(G) \subset \Omega$, and $g : \Omega \rightarrow \mathbb{C}$ be an analytic function with $g'(w) \neq 0$ for any $w \in \Omega$. Show that f is analytic if

$$h(z) = g(f(z)), \quad \text{for all } z \in G.$$

Source: August 2024, I.1

Exercise 3.15.2. Let G be a region, and

$$G^* = \{z : \bar{z} \in G\}.$$

If $f : G \rightarrow \mathbb{C}$ is analytic, prove that $f^* : G^* \rightarrow \mathbb{C}$ defined by

$$f^*(z) = \overline{f(\bar{z})}$$

is also analytic.

Source: August 2024, I.5

Exercise 3.15.3. Let f be an entire function, taking real values on the real axis, and imaginary values on the imaginary axis. Show that $f(z)$ is an odd function; that is,

$$f(-z) = -f(z)$$

for all z .

Source: August 2020, I.5

Exercise 3.15.4. Let

$$f(z) = e^x \sin y + ie^x \cos y, \quad z = x + iy.$$

Determine whether f is analytic on the complex plane.

Source: Fall 2025, I.5

Exercise 3.15.5. Let f be entire, injective, and without fixed points in $\mathbb{C}$. Show that

$$f(z) = az + b, \quad z \in \mathbb{C},$$

for some $a, b \in \mathbb{C}$ with $a \neq 0$.

Source: August 2022, I.4

3.16 Open mapping and restricted ranges

Exercise 3.16.1. Let G be a region, and suppose that $f: G \rightarrow \mathbb{C}$ is analytic such that $f(G)$ is a subset of a circle. Show that f is constant.

Source: Fall 2025, I.4

Exercise 3.16.2. Determine, with proof, all entire functions f such that for all $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and all $z \in \mathbb{C}$ the equation

$$f(\lambda z) = \lambda f(z)$$

holds.

Source: August 2018, I.4

Exercise 3.16.3. Prove that if f is a non-constant bounded holomorphic function defined on the half plane

$$\{z : \operatorname{Im} z < 1\},$$

then there is some point on the real axis where the value of f is not real.

Source: August 2018, II.3

3.17 Harmonic functions

Exercise 3.17.1. Prove that a nonconstant harmonic function on a region is an open map.

Source: August 2024, II.1

Exercise 3.17.2. Let G be a domain in the complex plane and $u: G \rightarrow \mathbb{R}$ be a nonconstant harmonic function and let

$$A = \{z \in G : u_x(z) = u_y(z) = 0\}.$$

Show that A does not have any limit points in G .

Source: January 2025, II.2

Exercise 3.17.3. If f is analytic on G and $f(z) \neq 0$ for any z show that

$$u = \log |f|$$

is harmonic on G .

Source: text exercise

Exercise 3.17.4. Let u be harmonic in G and suppose $\overline{B}(a; R) \subseteq G$. Show that

$$u(a) = \frac{1}{\pi R^2} \iint_{\overline{B}(a; R)} u(x, y) \, dx \, dy.$$

Source: text exercise

Advanced Coverage

4.1 Infinite-product functional equations

Exercise 4.1.1. Determine all entire functions that satisfy the functional equation

$$f(3z) = (1 - 3z)f(z).$$

Express the answer in terms of an infinite product.

Source: August 2018, II.1

4.2 Local geometry at a double zero

Exercise 4.2.1. Let $F(z)$ be holomorphic on a disc $D(z_0, r)$ so that

$$F(z_0) = F'(z_0) = 0 \quad \text{and} \quad F''(z_0) \neq 0.$$

Show that there are two curves Γ_1 and Γ_2 passing through z_0 that are orthogonal at z_0 and so that the function F when restricted to Γ_1 is real-valued and has a minimum at z_0 , and when restricted to Γ_2 it is also real-valued and has a maximum at z_0 .

Source: August 2018, II.2

4.3 Polynomial approximation versus compact convergence

Exercise 4.3.1. Show that there exists a sequence of polynomials (P_n) so that

$$\lim_{n \rightarrow \infty} P_n(z) = \begin{cases} 1, & \text{if } \operatorname{Re}(z) > 0, \\ 0, & \text{if } \operatorname{Re}(z) = 0, \\ -1, & \text{if } \operatorname{Re}(z) < 0. \end{cases}$$

Also, explain why (P_n) can't be bounded on compact subsets of $\mathbb{C}$.

Source: August 2018, II.4

Exercise 4.3.2. Prove or disprove: There exists a sequence of polynomials that converges pointwise on $\mathbb{C}$ to

$$\delta(z) = \begin{cases} 1, & z = 0, \\ 0, & z \neq 0. \end{cases}$$

Source: August 2022, II.3

4.4 Laurent series and isolated singularities

Exercise 4.4.1. Let

$$f(z) = \frac{1}{z(z-1)(z-2)}.$$

Give the Laurent expansion of $f(z)$ in each of the following annuli:

- (a) $\text{ann}(0; 0, 1)$;
- (b) $\text{ann}(0; 1, 2)$;
- (c) $\text{ann}(0; 2, \infty)$. *Source: text exercise*

Exercise 4.4.2. Let f be analytic in

$$G = \{z : 0 < |z - a| < r\}$$

except that there is a sequence of poles $\{a_n\}$ in G with $a_n \rightarrow a$. Show that for any $\omega \in \mathbb{C}$ there exists a sequence $\{z_n\} \subset G$ with

$$a = \lim z_n \quad \text{and} \quad \omega = \lim f(z_n).$$

Source: text exercise

4.5 Residues for real integrals

Exercise 4.5.1. Verify the following equation:

$$\int_0^\infty \frac{\cos(ax)}{(1+x^2)^2} dx = \frac{\pi(a+1)e^{-a}}{4}, \quad a > 0.$$

Source: text exercise

4.6 Boundary modulus and finite Blaschke products

Exercise 4.6.1. Suppose f is analytic in some region containing $\overline{B}(0, 1)$ and $|f(z)| = 1$ where $|z| = 1$. Find a formula for f . Hint: First consider the case where f has no zeros in $B(0, 1)$.

Source: text exercise

Exercise 4.6.2. Suppose that both f and g are analytic on $\overline{B(0; R)}$ with

$$|f(z)| = |g(z)| \quad \text{for } |z| = R.$$

Show that if neither f nor g vanishes in $B(0; R)$, then there is a constant λ , with $|\lambda| = 1$, such that $f = \lambda g$.

Source: text exercise

Exercise 4.6.3. Let f be analytic on $\mathbb{D}$, continuous on $\overline{\mathbb{D}}$, and satisfy $|f| = 1$ on $\partial\mathbb{D}$. Find a formula for f . *Hint: Consider first the zero-free case.*

Source: August 2022, II.4

4.7 Riemann mapping and topology of punctures

Exercise 4.7.1. (1) Let G be a region, let $a \in G$ and suppose that $f : (G \setminus \{a\}) \rightarrow \mathbb{C}$ is an analytic function such that $f(G \setminus \{a\}) = \Omega$ is bounded. Show that f has a removable singularity at $z = a$. If f is one-one, show that $f(a) \in \partial\Omega$.

(2) Show that there is no one-one analytic function which maps

$$G = \{z : 0 < |z| < 1\}$$

onto an annulus

$$\Omega = \{z : r < |z| < R\}$$

where $r > 0$. *Source: text exercise*

Exercise 4.7.2. Let G be a simply connected region which is not the whole plane and suppose that $\bar{z} \in G$ whenever $z \in G$. Let $a \in G \cap \mathbb{R}$ and suppose that $f : G \rightarrow D = \{z : |z| < 1\}$ is a one-one analytic function with $f(a) = 0$, $f'(a) > 0$ and $f(G) = D$. Let

$$G_+ = \{z \in G : \Im z > 0\}.$$

Show that $f(G_+)$ must lie entirely above or entirely below the real axis.

Source: text exercise

4.8 Subharmonic functions

Exercise 4.8.1. Let G and Ω be domains in the complex plane. If $f : G \rightarrow \Omega$ is one-to-one and analytic and $\varphi : \Omega \rightarrow \mathbb{R}$ is subharmonic, show that $\varphi \circ f$ is subharmonic.

Source: January 2025, II.1

4.9 Runge and approximation obstructions

Exercise 4.9.1. Let G be the open unit disk $B(0; 1)$ and let $K = \{z : \frac{1}{4} \leq |z| \leq \frac{3}{4}\}$. Show that there is a function f analytic on some open subset G_1 containing K which cannot be approximated on K by functions in $H(G)$.

Source: text exercise

4.10 Phragmén–Lindelöf

Exercise 4.10.1. Let $G = \{z : |\Im z| < \frac{1}{2}\pi\}$ and suppose $f : G \rightarrow \mathbb{C}$ is analytic and

$$\limsup_{z \rightarrow w} |f(z)| \leq M \quad \text{for } w \in \partial G.$$

Also, suppose $A < \infty$ and $a < 1$ can be found such that

$$|f(z)| < \exp[A \exp(a \Re z)]$$

for all z in G . Show that $|f(z)| \leq M$ for all z in G . Examine $\exp(\exp z)$ to see that this is the best possible growth condition. Can we take $a = 1$ above?

Source: text exercise

Exercise 4.10.2. Let $f : G \rightarrow \mathbb{C}$ be analytic and suppose that G is bounded. Fix $z_0 \in \partial G$ and suppose that

$$\limsup_{z \rightarrow w} |f(z)| \leq M \quad \text{for } w \in \partial G, w \neq z_0.$$

Show that if

$$\lim_{z \rightarrow z_0} |z - z_0|^\varepsilon |f(z)| = 0$$

for every $\varepsilon > 0$ then $|f(z)| \leq M$ for every $z \in G$. *Hint:* If $a \notin G$, consider $\varphi(z) = (z - z_0)(z - a)^{-1}$.

Source: text exercise

Warm-ups and Supporting Exercises

Use these for speed, but do not spend your main study time on them before completing Tier 1:

- Calculate the following:
 - (1) the square roots of i
 - (2) the cube roots of i
 - (3) the square roots of $\sqrt{3} + 3i$ *Source: Exercise/complex.tex, line 112*
- If (X, d) is any metric space show that every open ball is an open set. Also, show that every closed ball is a closed set.

Source: Exercise/complex.tex, line 269
- Let γ and σ be the two polygons $[1, i]$ and $[1, 1 + i, i]$. Express γ and σ as paths and calculate $\int_{\gamma} f$ and $\int_{\sigma} f$ where $f(z) = |z|^2$.

Source: Exercise/complex.tex, line 1588
- Find the radius of convergence for each of the following power series:
 - (1) $\sum_{n=0}^{\infty} a^n z^n, \quad a \in \mathbb{C};$
 - (2) $\sum_{n=0}^{\infty} a^{n^2} z^n, \quad a \in \mathbb{C};$
 - (3) $\sum_{n=0}^{\infty} k^n z^n, \quad k \text{ an integer } \neq 0;$
 - (4) $\sum_{n=0}^{\infty} z^{n!}$. *Source: Exercise/complex.tex, line 860*
- Show that if F_1 and F_2 are primitives for $f : G \rightarrow \mathbb{C}$ and G is open and connected, then there is a constant c such that $F_1(z) = c + F_2(z)$ for each $z \in G$.

Source: Exercise/complex.tex, line 1747

Recommended Study Plan

1. First pass: items 1–9 (core first-semester techniques).
2. Second pass: items 10–17 (convergence, factorization, and harmonic theory).
3. Third pass: items 18–27 (Part II and advanced topics).
4. Final review: redo every qualifying-exam question above under a 20–25 minute limit without notes.

A solution counts as exam-ready only if you can state exactly why the relevant hypotheses hold—for example, why a quotient has removable singularities, why a logarithm exists on the domain, which zeros lie inside a Rouché contour, or why a family is locally bounded.

Part II

Theorem Reference

Topology, compactness, and uniform convergence

The metric-space results repeatedly used to justify connectedness, compactness, distance estimates, and passage to limits.

7.1 Chapter II, §2 — Connectedness

Definition 7.1.1 (Connected set). *Conway II.2.1; printed p. 14; Definition.*

A metric space (X, d) is connected if the only subsets of X which are both open and closed are $\emptyset$ and X . If $A \subset X$ then A is a connected subset of X if the metric space (A, d) is connected.

Used for. Regions, components, identity arguments, and open mapping.

Theorem 7.1.2 (Polygonal characterization of regions). *Conway II.2.3; printed p. 15; Important.*

An open set $G \subset \mathbb{C}$ is connected iff for any two points a, b in G there is a polygon from a to b lying entirely inside G .

Used for. Contour construction, regions, primitives, and connected-domain arguments.

Corollary 7.1.3 (Axis-parallel polygonal paths). *Conway II.2.4; printed p. 15; Important.*

If $G \subset \mathbb{C}$ is open and connected and a and b are points in G then there is a polygon P in G from a to b which is made up of line segments parallel to either the real or imaginary axis.

Used for. Cauchy–Riemann and path-integration arguments.

Lemma 7.1.4 (Union of connected sets). *Conway II.2.6; printed p. 16; Supporting.*

Let $x_0 \in X$ and let $\{D_j : j \in J\}$ be a collection of connected subsets of X such that $x_0 \in D_j$ for each j in J . Then $D = \bigcup \{D_j : j \in J\}$ is connected.

Used for. Connectedness exercises and component arguments.

7.2 Chapter II, §3 — Sequences and completeness

Theorem 7.2.1 (Cantor intersection theorem). *Conway II.3.7; printed p. 19; Supporting.*

A metric space (X, d) is complete iff for any sequence $\{F_n\}$ of non-empty closed sets with $F_1 \supset F_2 \supset \cdots$ and $\text{diam } F_n \rightarrow 0$, $\bigcap_{n=1}^{\infty} F_n$ consists of a single point.

Used for. Completeness exercises and nested-disk arguments.

7.3 Chapter II, §4 — Compactness

Definition 7.3.1 (Compact set). *Conway II.4.1; printed p. 20; Definition.*

A subset K of a metric space X is compact if for every collection $\mathcal{G}$ of open sets in X with the property $K \subset \bigcup\{G : G \in \mathcal{G}\}$, there are a finite number of sets $G_1, \dots, G_n$ in $\mathcal{G}$ such that $K \subset G_1 \cup G_2 \cup \cdots \cup G_n$. A collection of sets $\mathcal{G}$ satisfying the displayed inclusion is called a cover of K ; if each member of $\mathcal{G}$ is an open set it is called an open cover of K .

Used for. Finite subcovers, uniform estimates, extrema, and compact convergence.

Lemma 7.3.2 (Lebesgue covering lemma). *Conway II.4.8; printed p. 21; Supporting.*

If (X, d) is sequentially compact and $\mathcal{G}$ is an open cover of X then there is an $\varepsilon > 0$ such that if x is in X , there is a set G in $\mathcal{G}$ with $B(x; \varepsilon) \subset G$.

Used for. Uniform local estimates on compact curves and sets.

Theorem 7.3.3 (Equivalent forms of compactness). *Conway II.4.9; printed p. 22; Important.*

Let (X, d) be a metric space; then the following are equivalent statements:

- (a) X is compact
- (b) every infinite set in X has a limit point
- (c) X is sequentially compact
- (d) X is complete and for every $\varepsilon > 0$ there are a finite number of points $x_1, \dots, x_n$ in X such that $X = \bigcup_{k=1}^n B(x_k; \varepsilon)$.

Used for. Normal families, compact exhaustion, and topology exercises.

Theorem 7.3.4 (Heine–Borel theorem). *Conway II.4.10; printed p. 23; Important.*

A subset K of $\mathbb{R}^n$ ($n \geq 1$) is compact iff K is closed and bounded.

Used for. Compact contours, boundary extrema, and finite subcovers.

7.4 Chapter II, §5 — Continuity

Theorem 7.4.1 (Continuous images). *Conway II.5.8; printed p. 26; Important.*

Let $f : (X, d) \rightarrow (\Omega, \rho)$ be a continuous function.

- (a) If X is compact then $f(X)$ is a compact subset of Ω .
- (b) If X is connected then $f(X)$ is a connected subset of Ω .

Used for. Range restrictions, integer-valued continuous functions, and extrema.

Corollary 7.4.2 (Intermediate-value property). *Conway II.5.10; printed p. 26; Supporting.*

If $f : X \rightarrow \mathbb{R}$ is continuous and X is connected then $f(X)$ is an interval.

Used for. Harmonic open-map and sign arguments.

Corollary 7.4.3 (Extreme-value theorem). *Conway II.5.12; printed p. 27; Important.*

If $f : X \rightarrow \mathbb{R}$ is continuous and $K \subset X$ is compact then there are points x_0 and y_0 in K with $f(x_0) = \sup\{f(x) : x \in K\}$ and $f(y_0) = \inf\{f(x) : x \in K\}$.

Used for. Minimum-modulus, contour-distance, and compact-boundary arguments.

Theorem 7.4.4 (Heine–Cantor theorem). *Conway II.5.15; printed p. 27; Supporting.*

Suppose $f : X \rightarrow \Omega$ is continuous and X is compact; then f is uniformly continuous.

Used for. Moving-point convergence and uniform control on curves.

Theorem 7.4.5 (Positive separation from a compact set). *Conway II.5.17; printed p. 28; Supporting.*

If A and B are non-empty disjoint sets in X with B closed and A compact then $d(A, B) > 0$.

Used for. Choosing small contours and compact subsets away from boundaries.

7.5 Chapter II, §6 — Uniform convergence

Theorem 7.5.1 (Uniform limit theorem). *Conway II.6.1; printed p. 29; Important.*

Suppose $f_n : (X, d) \rightarrow (\Omega, \rho)$ is continuous for each n and that $f = u\text{-}\lim f_n$; then f is continuous.

Used for. Continuity of limits and convergence in function spaces.

Theorem 7.5.2 (Weierstrass M-test). *Conway II.6.2; printed p. 29; Important.*

Let $u_n : X \rightarrow \mathbb{C}$ be a function such that $|u_n(x)| \leq M_n$ for every x in X and suppose the constants satisfy $\sum_{n=1}^{\infty} M_n < \infty$. Then $\sum_1^{\infty} u_n$ is uniformly convergent.

Used for. Power series, infinite products, Mittag-Leffler, and Weierstrass constructions.

7.6 Chapter III, §2 — Analytic functions

Definition 7.6.1 (Region). *Conway III.§2; printed p. 41; Definition.*

A region is an open connected subset of the plane.

Used for. Reading nearly every global theorem hypothesis in the reference.

Analytic functions, branches, and mappings

Local analytic structure, Cauchy–Riemann theory, logarithms, and Möbius transformations.

8.1 Chapter III, §1 — Power series

Theorem 8.1.1 (Radius of convergence theorem). *Conway III.1.3; printed p. 31; Important.*

For a given power series $\sum_{n=0}^{\infty} a_n(z-a)^n$ define the number R , $0 \leq R \leq \infty$, by $1/R = \limsup |a_n|^{1/n}$; then:

- (a) if $|z-a| < R$, the series converges absolutely
- (b) if $|z-a| > R$, the terms of the series become unbounded and so the series diverges
- (c) if $0 < r < R$ then the series converges uniformly on $\{z : |z-a| \leq r\}$. Moreover, the number R is the only number having properties (a) and (b).

Used for. All power-series radius and boundary-convergence problems.

Proposition 8.1.2 (Ratio formula for the radius of convergence). *Conway III.1.4; printed p. 32; Important.*

If $\sum_{n=0}^{\infty} a_n(z-a)^n$ is a given power series with radius of convergence R , then

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

if this limit exists.

Used for. Computing a power series' radius of convergence from consecutive coefficients.

Formula 8.1.3 (Exponential series). *Conway III.§1; printed p. 32; Formula.*

The series $\sum_{n=0}^{\infty} \frac{z^n}{n!}$ has radius of convergence ∞ , so it converges at every complex number and uniformly on every compact subset of $\mathbb{C}$. The exponential series or function is designated by

$$e^z = \exp z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

Used for. Power-series arguments involving e^z , entire functions, derivatives, and exponential equations.

8.2 Chapter III, §2 — Analytic functions

Definition 8.2.1 (Complex differentiability). *Conway III.2.1; printed p. 33; Definition.*

If G is an open set in $\mathbb{C}$ and $f : G \rightarrow \mathbb{C}$ then f is differentiable at a point a in G if $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists; the value of this limit is denoted by $f'(a)$ and is called the derivative of f at a . If f is differentiable at each point of G we say that f is differentiable on G .

Used for. Checking whether a function is complex differentiable.

Definition 8.2.2 (Analytic function). *Conway III.2.3; printed p. 34; Definition.*

A function $f : G \rightarrow \mathbb{C}$ is analytic if f is continuously differentiable on G .

Used for. The basic hypothesis for Cauchy theory and all later analytic results.

Corollary 8.2.3 (Analyticity of a power series). *Conway III.2.9; printed p. 37; Important.*

If the series $\sum_{n=0}^{\infty} a_n(z-a)^n$ has radius of convergence $R > 0$ then $f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n$ is analytic in $B(a; R)$.

Used for. Taylor expansions, functional equations, and coefficient comparisons.

Proposition 8.2.4 (Zero derivative implies constant). *Conway III.2.10; printed p. 37; Important.*

If G is open and connected and $f : G \rightarrow \mathbb{C}$ is differentiable with $f'(z) = 0$ for all z in G , then f is constant.

Used for. Uniqueness of primitives and constant-function conclusions.

Definition 8.2.5 (Branch of the logarithm). *Conway III.2.18; printed p. 39; Definition.*

If G is an open connected set in $\mathbb{C}$ and $f : G \rightarrow \mathbb{C}$ is a continuous function such that $z = \exp f(z)$ for all z in G then f is a branch of the logarithm.

Used for. Logarithm branches, complex powers, roots, and contour obstructions.

Proposition 8.2.6 (Uniqueness of logarithm branches). *Conway III.2.19; printed p. 39; Important.*

If $G \subset \mathbb{C}$ is open and connected and f is a branch of $\log z$ on G then the totality of branches of $\log z$ are the functions $f(z) + 2\pi ki$, $k \in \mathbb{Z}$.

Used for. Branch classification and annulus problems.

Proposition 8.2.7 (Analytic inverse criterion). *Conway III.2.20; printed p. 39; Important.*

Let G and Ω be open subsets of $\mathbb{C}$. Suppose that $f : G \rightarrow \mathbb{C}$ and $g : \Omega \rightarrow \mathbb{C}$ are continuous functions such that $f(G) \subset \Omega$ and $g(f(z)) = z$ for all z in G . If g is differentiable and $g'(z) \neq 0$, f is differentiable and $f'(z) = 1/g'(f(z))$. If g is analytic, f is analytic.

Used for. August 2024 I.1 and analytic inverses.

Corollary 8.2.8 (Analyticity of logarithm branches). *Conway III.2.21; printed p. 40; Important.*

A branch of the logarithm function is analytic and its derivative is z^{-1} .

Used for. Principal branches, powers, and path integrals of $1/z$.

Definition 8.2.9 (Branches of complex powers). *Conway III.2 — Complex powers; printed p. 40; Definition.*

Let G be open and connected, let f be a branch of the logarithm on G , and fix $b \in \mathbb{C}$. A branch of z^b on G is defined by

$$z^b = \exp(bf(z)).$$

If $b \in \mathbb{Z}$, this agrees with the ordinary integer power and is independent of the logarithm branch. When no branch is specified, z^b means

$$z^b = \exp(b \operatorname{Log} z),$$

where Log is the principal logarithm. Every such branch of z^b is analytic on its domain.

Used for. Analytic powers and roots, branch dependence, August 2020 II.5, and Fall 2025 I.7.

Formula 8.2.10 (Cauchy–Riemann equations). *Conway III.2.24; printed p. 41; Formula.*

If $f(x + iy) = u(x, y) + iv(x, y)$, the Cauchy–Riemann equations are

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Used for. Direct analyticity tests, harmonic conjugates, and computations of derivatives.

Remark 8.2.11 (Differentiability implies Cauchy–Riemann). *Conway III.2.24 — note; printed p. 41; Remark.*

Let $z_0 = x_0 + iy_0$ and write $f = u + iv$. If f is complex differentiable at z_0 , then the first partial derivatives of u and v at (x_0, y_0) exist and

$$u_x(x_0, y_0) = v_y(x_0, y_0), \quad u_y(x_0, y_0) = -v_x(x_0, y_0).$$

This implication is necessary only: satisfying the Cauchy–Riemann equations at a single point does not by itself imply that f is complex differentiable there.

Used for. Using Cauchy–Riemann to rule out differentiability and distinguishing necessary from sufficient conditions.

Theorem 8.2.12 (Cauchy–Riemann criterion). *Conway III.2.29; printed p. 42; Essential.*

Let u and v be real-valued functions defined on a region G and suppose that u and v have continuous partial derivatives. Then $f : G \rightarrow \mathbb{C}$ defined by $f(z) = u(z) + iv(z)$ is analytic iff u and v satisfy the Cauchy–Riemann equations.

Used for. Direct analyticity checks, reflection, and harmonic conjugates.

Theorem 8.2.13 (Existence of harmonic conjugates on disks). *Conway III.2.30; printed p. 43; Important.*

Let G be either the whole plane $\mathbb{C}$ or some open disk. If $u : G \rightarrow \mathbb{R}$ is a harmonic function then u has a harmonic conjugate.

Used for. Local harmonic arguments and harmonic open mapping.

8.3 Chapter III, §3 — Analytic mappings and Möbius transformations

Theorem 8.3.1 (Angle preservation). *Conway III.3.4; printed p. 46; Supporting.*

If $f : G \rightarrow \mathbb{C}$ is analytic then f preserves angles at each point z_0 of G where $f'(z_0) \neq 0$.

Used for. Conformal-map verification and local geometry.

Theorem 8.3.2 (Möbius maps preserve generalized circles). *Conway III.3.14; printed p. 49; Important.*

A Möbius transformation takes circles onto circles.

Used for. Disk/half-plane maps, slit domains, fixed points, and circle geometry.

Complex integration, Cauchy theory, and zeros

The central toolkit for contour computation, estimates, identities, and local mapping behavior.

9.1 Chapter IV, §1 — Riemann–Stieltjes integrals

Theorem 9.1.1 (Parametrization formula). *Conway IV.1.9; printed p. 62; Supporting.*

If γ is piecewise smooth and $f : [a, b] \rightarrow \mathbb{C}$ is continuous then $\int_a^b f d\gamma = \int_a^b f(t)\gamma'(t) dt$.

Used for. Direct evaluation of contour integrals.

Proposition 9.1.2 (Basic properties and estimate for line integrals). *Conway IV.1.17; printed p. 65; Important.*

Let γ be a rectifiable curve and suppose that f is a function continuous on $\{\gamma\}$. Then:

(a) $\int_{\gamma} f = -\int_{-\gamma} f$

(b)

$$\left| \int_{\gamma} f \right| \leq \int_{\gamma} |f| |dz| \leq V(\gamma) \sup\{|f(z)| : z \in \{\gamma\}\}$$

(c) if $c \in \mathbb{C}$ then $\int_{\gamma} f(z) dz = \int_{\gamma+c} f(z-c) dz$.

Used for. Large-arc limits, Liouville contour proofs, and vanishing integrals.

Definition 9.1.3 (Primitive). *Conway IV.§1; printed p. 65; Definition.*

F is a primitive of f when $F' = f$.

Used for. The fundamental theorem for line integrals, path independence, and logarithms.

Theorem 9.1.4 (Fundamental theorem for complex line integrals). *Conway IV.1.18; printed p. 65; Important.*

Let G be open in $\mathbb{C}$ and let γ be a rectifiable path in G with initial and end points α and β respectively. If $f : G \rightarrow \mathbb{C}$ is a continuous function with a primitive $F : G \rightarrow \mathbb{C}$, then $\int_{\gamma} f = F(\beta) - F(\alpha)$.

Used for. Primitives, path independence, and integrals of powers.

Corollary 9.1.5 (Integral of a primitive over a closed curve). *Conway IV.1.22; printed p. 66; Supporting.*

Let G , γ , and f satisfy the same hypothesis as in Theorem 1.18. If γ is a closed curve then $\int_{\gamma} f = 0$.

Used for. Primitives, path independence, and logarithms.

9.2 Chapter IV, §2 — Power-series representation

Proposition 9.2.1 (Differentiation under the integral sign). *Conway IV.2.1; printed p. 68; Supporting.*

Let $\varphi : [a, b] \times [c, d] \rightarrow \mathbb{C}$ be a continuous function and define $g : [c, d] \rightarrow \mathbb{C}$ by $g(t) = \int_a^b \varphi(s, t) ds$. Then g is continuous. Moreover, if $\partial\varphi/\partial t$ exists and is a continuous function on $[a, b] \times [c, d]$ then g is continuously differentiable and $g'(t) = \int_a^b \frac{\partial\varphi}{\partial t}(s, t) ds$.

Used for. Differentiation under the integral sign and harmonic conjugates.

Lemma 9.2.2 (Uniform convergence under contour integrals). *Conway IV.2.7; printed p. 71; Important.*

Let γ be a rectifiable curve in $\mathbb{C}$ and suppose that F_n and F are continuous functions on $\{\gamma\}$. If $F = u\text{-}\lim F_n$ on $\{\gamma\}$ then $\int_{\gamma} F = \lim \int_{\gamma} F_n$.

Used for. Analytic limits, Laurent series, and contour approximations.

Theorem 9.2.3 (Local power-series representation). *Conway IV.2.8; printed p. 72; Essential.*

Let f be analytic in $B(a; R)$; then $f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n$ for $|z-a| < R$, where $a_n = \frac{1}{n!} f^{(n)}(a)$, and this series has radius of convergence $\geq R$.

Used for. Taylor expansions, symmetry, and classification of entire functions.

Corollary 9.2.4 (Taylor expansion on the largest centered disk). *Conway IV.2.11; printed p. 73; Important.*

If $f : G \rightarrow \mathbb{C}$ is analytic and $a \in G$ then $f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n$ for $|z-a| < R$, where $R = d(a, \partial G)$.

Used for. Taylor expansions and radii of convergence.

Corollary 9.2.5 (Infinite differentiability). *Conway IV.2.12; printed p. 73; Supporting.*

If $f : G \rightarrow \mathbb{C}$ is analytic then f is infinitely differentiable.

Used for. Derivative arguments and analytic regularity.

Corollary 9.2.6 (Cauchy formula for derivatives). *Conway IV.2.13; printed p. 73; Essential.*

If $f : G \rightarrow \mathbb{C}$ is analytic and $\overline{B(a; r)} \subset G$ then

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w-a)^{n+1}} dw$$

, where $\gamma(t) = a + re^{it}$, $0 \leq t \leq 2\pi$.

Used for. High-order pole integrals and coefficient formulas.

Theorem 9.2.7 (Cauchy estimates). *Conway IV.2.14; printed p. 74; Essential.*

Let f be analytic in $B(a; R)$ and suppose $|f(z)| \leq M$ for all z in $B(a; R)$. Then $|f^{(n)}(a)| \leq n!M/R^n$.

Used for. Liouville, polynomial-growth results, derivative bounds, and normal families.

Remark 9.2.8 (Boundary form of the Cauchy estimates). *Conway IV.2.14 — boundary form; printed p. 74; Remark.*

Let f be analytic in $B(a; R)$. For every $0 < r < R$, combining Cauchy's formula with the later maximum modulus theorem gives

$$|f^{(n)}(a)| \leq \frac{n!}{r^n} \max_{|z-a|=r} |f(z)|.$$

Used for. Sharper derivative bounds using the later maximum modulus theorem.

9.3 Chapter IV, §3 — Zeros of an analytic function

Definition 9.3.1 (Zero and multiplicity). *Conway IV.3.1; printed p. 76; Definition.*

If $f : G \rightarrow \mathbb{C}$ is analytic and a in G satisfies $f(a) = 0$ then a is a zero of f of multiplicity $m \geq 1$ if there is an analytic function $g : G \rightarrow \mathbb{C}$ such that $f(z) = (z-a)^m g(z)$ where $g(a) \neq 0$.

Used for. Counting zeros correctly in factorization, residues, and Rouché arguments.

Definition 9.3.2 (Entire function). *Conway IV.3.2; printed p. 77; Definition.*

An entire function is a function which is defined and analytic in the whole complex plane $\mathbb{C}$.

Used for. Liouville, polynomial growth, factorization, and Picard problems.

Theorem 9.3.3 (Liouville's theorem). *Conway IV.3.4; printed p. 77; Essential.*

If f is a bounded entire function then f is constant.

Used for. FTA proofs, bounded ratios, harmonic Liouville, and entire growth.

Theorem 9.3.4 (Fundamental theorem of algebra). *Conway IV.3.5; printed p. 77; Essential.*

If $p(z)$ is a non constant polynomial then there is a complex number a with $p(a) = 0$.

Used for. FTA questions and polynomial zero counts.

Theorem 9.3.5 (Identity theorem). *Conway IV.3.7; printed p. 78; Essential.*

Let G be a connected open set and let $f : G \rightarrow \mathbb{C}$ be an analytic function. Then the following are equivalent statements:

(a) $f \equiv 0$

(b) there is a point a in G such that $f^{(n)}(a) = 0$ for each $n \geq 0$

(c) $\{z \in G : f(z) = 0\}$ has a limit point in G .

Used for. Reflection identities, zero sets, functional equations, and uniqueness.

Corollary 9.3.6 (Identity theorem for two functions). *Conway IV.3.8; printed p. 79; Important.*

If f and g are analytic on a region G then $f \equiv g$ iff $\{z \in G : f(z) = g(z)\}$ has a limit point in G .

Used for. Reflection identities, functional equations, and uniqueness.

Corollary 9.3.7 (Finite multiplicity of zeros). *Conway IV.3.9; printed p. 79; Important.*

If f is analytic on an open connected set G and f is not identically zero then for each a in G with $f(a) = 0$ there is an integer $n \geq 1$ and an analytic function $g : G \rightarrow \mathbb{C}$ such that $g(a) \neq 0$ and $f(z) = (z - a)^n g(z)$ for all z in G . That is, each zero of f has finite multiplicity.

Used for. Quotient removal and prescribed multiplicities.

Corollary 9.3.8 (Isolated zeros). *Conway IV.3.10; printed p. 79; Important.*

If $f : G \rightarrow \mathbb{C}$ is analytic and not constant, $a \in G$, and $f(a) = 0$ then there is an $R > 0$ such that $B(a; R) \subset G$ and $f(z) \neq 0$ for $0 < |z - a| < R$.

Used for. Critical sets, local factorization, and zero counting.

9.4 Chapter IV, §4 — Index of a closed curve

Definition 9.4.1 (Index (winding number)). *Conway IV.4.2; printed p. 81; Definition.*

If γ is a closed rectifiable curve in $\mathbb{C}$ then for $a \notin \{\gamma\}$,

$$n(\gamma; a) = \frac{1}{2\pi i} \int_{\gamma} (z - a)^{-1} dz$$

is called the index of γ with respect to the point a . It is also sometimes called the winding number of γ around a .

Used for. Cauchy formulas, residues, and contours that are not simple circles.

Theorem 9.4.2 (Index of a closed curve). *Conway IV.4.4; printed p. 82; Important.*

Let γ be a closed rectifiable curve in $\mathbb{C}$. Then $n(\gamma; a)$ is constant for a belonging to a component of $G = \mathbb{C} - \{\gamma\}$. Also, $n(\gamma; a) = 0$ for a belonging to the unbounded component of G .

Used for. Winding-number integrals, logarithm obstruction, and noncircular contours.

9.5 Chapter IV, §5 — Cauchy's theorem and integral formula

Theorem 9.5.1 (Cauchy's integral formula — first version). *Conway IV.5.4; printed p. 84; Essential.*

Let G be an open subset of the plane and $f : G \rightarrow \mathbb{C}$ an analytic function. If γ is a closed rectifiable curve in G such that $n(\gamma; w) = 0$ for all w in $\mathbb{C} - G$, then for a in $G - \{\gamma\}$,

$$n(\gamma; a)f(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - a} dz$$

.

Used for. Contour integrals and local residue computation.

Theorem 9.5.2 (Cauchy's integral formula — second version). *Conway IV.5.6; printed p. 85; Important.*

Let G be an open subset of the plane and $f : G \rightarrow \mathbb{C}$ an analytic function. If $\gamma_1, \dots, \gamma_m$ are closed rectifiable curves in G such that $n(\gamma_1; w) + \dots + n(\gamma_m; w) = 0$ for all w in $\mathbb{C} - G$, then for a in $G - \bigcup_{j=1}^m \{\gamma_j\}$,

$$f(a) \sum_{k=1}^m n(\gamma_k; a) = \sum_{k=1}^m \frac{1}{2\pi i} \int_{\gamma_k} \frac{f(z)}{z - a} dz$$

.

Used for. Cauchy formulas on annuli and multiple curves.

Theorem 9.5.3 (Cauchy's theorem — first version). *Conway IV.5.7; printed p. 85; Supporting.*

Let G be an open subset of the plane and $f : G \rightarrow \mathbb{C}$ an analytic function. If $\gamma_1, \dots, \gamma_m$ are closed rectifiable curves in G such that $n(\gamma_1; w) + \dots + n(\gamma_m; w) = 0$ for all w in $\mathbb{C} - G$ then $\sum_{k=1}^m \int_{\gamma_k} f = 0$.

Used for. General contour deformation and multiply connected regions.

Theorem 9.5.4 (Cauchy formula for derivatives — several curves). *Conway IV.5.8; printed p. 86; Supporting.*

Let G be an open subset of the plane and $f : G \rightarrow \mathbb{C}$ an analytic function. If $\gamma_1, \dots, \gamma_m$ are closed rectifiable curves in G such that $n(\gamma_1; w) + \dots + n(\gamma_m; w) = 0$ for all w in $\mathbb{C} - G$ then for a in $G - \{\gamma\}$ and $k \geq 1$,

$$f^{(k)}(a) \sum_{j=1}^m n(\gamma_j; a) = k! \sum_{j=1}^m \frac{1}{2\pi i} \int_{\gamma_j} \frac{f(z)}{(z-a)^{k+1}} dz$$

.

Used for. Higher derivatives and multiple contours.

Corollary 9.5.5 (Cauchy formula for derivatives — one curve). *Conway IV.5.9; printed p. 86; Important.*

Let G be an open set and $f : G \rightarrow \mathbb{C}$ an analytic function. If γ is a closed rectifiable curve in G such that $n(\gamma; w) = 0$ for all w in $\mathbb{C} - G$ then for a in $G - \{\gamma\}$,

$$f^{(k)}(a)n(\gamma; a) = \frac{k!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-a)^{k+1}} dz$$

.

Used for. High-order contour integrals.

Theorem 9.5.6 (Morera's theorem). *Conway IV.5.10; printed p. 86; Essential.*

Let G be a region and let $f : G \rightarrow \mathbb{C}$ be a continuous function such that $\int_T f = 0$ for every triangular path T in G ; then f is analytic in G .

Used for. Removable line segments and locally uniform limits.

9.6 Chapter IV, §6 — Homotopy and simple connectivity

Theorem 9.6.1 (Cauchy's theorem — second version). *Conway IV.6.6; printed p. 89; Important.*

If $f : G \rightarrow \mathbb{C}$ is an analytic function and γ is a closed rectifiable curve in G such that $\gamma \simeq 0$, then $\int_{\gamma} f = 0$.

Used for. Null-homotopic contours.

Theorem 9.6.2 (Cauchy's theorem — third version). *Conway IV.6.7; printed p. 90; Important.*

If γ_0 and γ_1 are two closed rectifiable curves in G and $\gamma_0 \simeq \gamma_1$ then $\int_{\gamma_0} f = \int_{\gamma_1} f$ for every function f analytic on G .

Used for. Deforming contours and evaluating integrals on complicated paths.

Definition 9.6.3 (Simply connected open set). *Conway IV.6.14; printed p. 93; Definition.*

An open set G is simply connected if G is connected and every closed curve in G is homotopic to zero.

Used for. Global primitives, logarithms, roots, and the Riemann mapping theorem.

Theorem 9.6.4 (Cauchy's theorem — fourth version). *Conway IV.6.15; printed p. 94; Essential.*

If G is simply connected then $\int_{\gamma} f = 0$ for every closed rectifiable curve γ and every analytic function f .

Used for. Cauchy theory on simply connected regions.

Corollary 9.6.5 (Existence of primitives). *Conway IV.6.16; printed p. 94; Essential.*

If G is simply connected and $f : G \rightarrow \mathbb{C}$ is analytic in G then f has a primitive in G .

Used for. Path independence, conformal mapping, and branches.

Corollary 9.6.6 (Analytic logarithm criterion). *Conway IV.6.17; printed p. 94; Essential.*

Let G be simply connected and let $f : G \rightarrow \mathbb{C}$ be an analytic function such that $f(z) \neq 0$ for any z in G . Then there is an analytic function $g : G \rightarrow \mathbb{C}$ such that $f(z) = \exp g(z)$. If $z_0 \in G$ and $e^{w_0} = f(z_0)$, we may choose g such that $g(z_0) = w_0$.

Used for. Equal-zero quotients, analytic roots, and factorization.

9.7 Chapter IV, §7 — Counting zeros and open mapping

Theorem 9.7.1 (Counting zeros by the logarithmic derivative). *Conway IV.7.2; printed p. 97; Supporting.*

Let G be a region and let f be an analytic function on G with zeros $a_1, \dots, a_m$ (repeated according to multiplicity). If γ is a closed rectifiable curve in G which does not pass through any point a_k and if $\gamma \simeq 0$ in G then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = \sum_{k=1}^m n(\gamma; a_k)$$

.

Used for. Polynomial degree and zero counts.

Corollary 9.7.2 (Counting solutions of an analytic equation). *Conway IV.7.3; printed p. 97; Supporting.*

Let f , G , and γ be as in the preceding theorem except that $a_1, \dots, a_m$ are the points in G that satisfy the equation $f(z) = \alpha$; then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z) - \alpha} dz = \sum_{k=1}^m n(\gamma; a_k)$$

.

Used for. Counting preimages and local integral formulas.

Theorem 9.7.3 (Local mapping theorem). *Conway IV.7.4; printed p. 98; Essential.*

Suppose f is analytic in $B(a; R)$ and let $\alpha = f(a)$. If $f(z) - \alpha$ has a zero of order m at $z = a$ then there is an $\varepsilon > 0$ and $\delta > 0$ such that for $0 < |\zeta - \alpha| < \delta$, the equation $f(z) = \zeta$ has exactly m simple roots in $B(a; \varepsilon)$.

Used for. Local saddle geometry, moving preimages, and open mapping.

Remark 9.7.4 (Local analytic inverse). *Conway IV.7.4 — consequence; printed p. 98; Remark.*

Let G be a region, let $f : G \rightarrow \mathbb{C}$ be analytic, and let $a \in G$. If $f'(a) \neq 0$, then there is an open neighborhood $U \subset G$ of a such that the restriction $f|_U : U \rightarrow f(U)$ is one-one and onto, $f(U)$ is open, and $(f|_U)^{-1} : f(U) \rightarrow U$ is analytic. For every $z \in U$,

$$\left((f|_U)^{-1}\right)'(f(z)) = \frac{1}{f'(z)}.$$

This is a local conclusion; f need not be one-one on all of G .

Used for. Local conformal coordinates, inverse branches, and distinguishing local from global injectivity.

Theorem 9.7.5 (Open mapping theorem). *Conway IV.7.5; printed p. 99; Essential.*

Let G be a region and suppose that f is a non constant analytic function on G . Then for any open set V in G , $f(V)$ is open.

Used for. Functions with image in a circle or line, harmonic open maps, and range arguments.

Corollary 9.7.6 (Global analytic inverse theorem). *Conway IV.7.6; printed p. 99; Important.*

Suppose G is a region and $f : G \rightarrow \mathbb{C}$ is one-one and analytic. Put $\Omega = f(G)$. Then Ω is open, $f'(z) \neq 0$ for every $z \in G$, and the global inverse $f^{-1} : \Omega \rightarrow G$ is analytic. If $w = f(z)$, then

$$(f^{-1})'(w) = \frac{1}{f'(z)}.$$

Used for. Conformal equivalence, inverse maps, and subharmonic pullbacks.

9.8 Chapter IV, §8 — Goursat's theorem

Theorem 9.8.1 (Goursat's theorem). *Conway IV.8; printed p. 100; Important.*

Let G be an open set and let $f : G \rightarrow \mathbb{C}$ be a differentiable function; then f is analytic on G .

Used for. Justifying analyticity under the minimal derivative definition.

Singularities, residues, and zero counting

Laurent classification, residue calculus, the argument principle, and Rouché's theorem.

10.1 Chapter V, §1 — Classification of singularities

Definition 10.1.1 (Isolated and removable singularities). *Conway V.1.1; printed p. 103; Definition.*

A function f has an isolated singularity at $z = a$ if there is an $R > 0$ such that f is defined and analytic in $B(a; R) - \{a\}$ but not in $B(a; R)$. The point a is called a removable singularity if there is an analytic function $g : B(a; R) \rightarrow \mathbb{C}$ such that $g(z) = f(z)$ for $0 < |z - a| < R$.

Used for. Classifying punctures and extending quotients across common zeros.

Theorem 10.1.2 (Removable-singularity criterion). *Conway V.1.2; printed p. 103; Essential.*

If f has an isolated singularity at a then the point $z = a$ is a removable singularity iff $\lim_{z \rightarrow a} (z - a)f(z) = 0$.

Used for. Punctured-domain maps, quotients at common zeros, and bounded extensions.

Definition 10.1.3 (Pole and essential singularity). *Conway V.1.3; printed p. 105; Definition.*

If $z = a$ is an isolated singularity of f then a is a pole of f if $\lim_{z \rightarrow a} |f(z)| = \infty$. That is, for any $M > 0$ there is a number $\varepsilon > 0$ such that $|f(z)| \geq M$ whenever $0 < |z - a| < \varepsilon$. If an isolated singularity is neither a pole nor a removable singularity it is called an essential singularity.

Used for. Distinguishing the three types of isolated singularity.

Proposition 10.1.4 (Representation at a pole). *Conway V.1.4; printed p. 105; Important.*

If G is a region with a in G and if f is analytic on $G - \{a\}$ with a pole at $z = a$ then there is a positive integer m and an analytic function $g : G \rightarrow \mathbb{C}$ such that $f(z) = \frac{g(z)}{(z - a)^m}$.

Used for. Pole order, singular parts, and residue computation.

Definition 10.1.5 (Order of a pole). *Conway V.1.6; printed p. 105; Definition.*

If f has a pole at $z = a$ and m is the smallest positive integer such that $f(z)(z - a)^m$ has a removable singularity at $z = a$ then f has a pole of order m at $z = a$.

Used for. Higher-order residue formulas and Laurent principal parts.

Development 10.1.6 (Laurent series development). *Conway V.1.11; printed p. 107; Essential.*

Let f be analytic in the annulus $\text{ann}(a; R_1, R_2)$. Then $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - a)^n$ where the convergence is absolute and uniform over $\overline{\text{ann}(a; r_1, r_2)}$ if $R_1 < r_1 < r_2 < R_2$. Also the coefficients a_n are given by the formula $a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{(z - a)^{n+1}} dz$, where γ is the circle $|z - a| = r$ for any r , $R_1 < r < R_2$. Moreover, this series is unique.

Used for. Laurent expansions, principal parts, residues, and singularity classification.

Corollary 10.1.7 (Classification by Laurent series). *Conway V.1.18; printed p. 109; Essential.*

Let $z = a$ be an isolated singularity of f and let $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - a)^n$ be its Laurent Expansion in $\text{ann}(a; 0, R)$. Then:

- (a) $z = a$ is a removable singularity iff $a_n = 0$ for $n \leq -1$
- (b) $z = a$ is a pole of order m iff $a_{-m} \neq 0$ and $a_n = 0$ for $n \leq -(m + 1)$
- (c) $z = a$ is an essential singularity iff $a_n \neq 0$ for infinitely many negative integers n .

Used for. Laurent expansions and singularity classification.

Theorem 10.1.8 (Casorati–Weierstrass theorem). *Conway V.1.21; printed p. 109; Important.*

If f has an essential singularity at $z = a$ then for every $\delta > 0$, $\{f[\text{ann}(a; 0, \delta)]\}^- = \mathbb{C}$.

Used for. Cluster-value problems and essential singularities.

10.2 Chapter V, §2 — Residues

Definition 10.2.1 (Residue). *Conway V.2.1; printed p. 112; Definition.*

Let f have an isolated singularity at $z = a$ and let $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - a)^n$ be its Laurent Expansion about $z = a$. Then the residue of f at $z = a$ is the coefficient a_{-1} . Denote this by $\text{Res}(f; a) = a_{-1}$.

Used for. Residue computation and contour integrals.

Theorem 10.2.2 (Residue theorem). *Conway V.2.2; printed p. 112; Essential.*

Let f be analytic in the region G except for the isolated singularities $a_1, a_2, \dots, a_m$. If γ is a closed rectifiable curve in G which does not pass through any of the points a_k and if $\gamma \simeq 0$ in G then

$$\frac{1}{2\pi i} \int_{\gamma} f = \sum_{k=1}^m n(\gamma; a_k) \operatorname{Res}(f; a_k)$$

.

Used for. All multi-pole contour integrals and real integrals by residues.

Proposition 10.2.3 (Residue at a pole). *Conway V.2.4; printed p. 113; Essential.*

Suppose f has a pole of order m at $z = a$ and put $g(z) = (z - a)^m f(z)$; then $\operatorname{Res}(f; a) = \frac{1}{(m-1)!} g^{(m-1)}(a)$.

Used for. High-order residues, including Fall 2025 Part II.

10.3 Chapter V, §3 — Argument principle

Definition 10.3.1 (Meromorphic function). *Conway V.3.3; printed p. 123; Definition.*

If G is open and f is a function defined and analytic in G except for poles, then f is a meromorphic function on G .

Used for. Residue theorem, argument principle, and Rouché's theorem.

Theorem 10.3.2 (Argument principle). *Conway V.3.4; printed p. 123; Essential.*

Let f be meromorphic in G with poles $p_1, p_2, \dots, p_m$ and zeros $z_1, z_2, \dots, z_n$ counted according to multiplicity. If γ is a closed rectifiable curve in G with $\gamma \simeq 0$ and not passing through $p_1, \dots, p_m; z_1, \dots, z_n$; then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = \sum_{k=1}^n n(\gamma; z_k) - \sum_{j=1}^m n(\gamma; p_j)$$

.

Used for. Counting solutions and change of argument.

Theorem 10.3.3 (Rouché's theorem). *Conway V.3.8; printed p. 125; Essential.*

Suppose f and g are meromorphic in a neighborhood of $\overline{B(a; R)}$ with no zeros or poles on the circle $\gamma = \{z : |z - a| = R\}$. If Z_f, Z_g (P_f, P_g) are the number of zeros (poles) of f and g inside γ counted according to their multiplicities and if $|f(z) + g(z)| < |f(z)| + |g(z)|$ on γ , then $Z_f - P_f = Z_g - P_g$.

Used for. All qualifying-exam root counts and Hurwitz-type arguments.

Corollary 10.3.4 (Classical Rouché inequality). *Conway V.3.8a; printed p. 126; Essential.*

Under the hypotheses of Rouché's Theorem, if $|f(z) + g(z)| < |g(z)|$ on γ , then $Z_f - P_f = Z_g - P_g$.

Used for. The usual comparison form used to count zeros of a perturbation.

Maximum principles, Schwarz lemma, and growth

Boundary estimates, disk automorphisms, three-circles convexity, and unbounded-domain maximum principles.

11.1 Chapter VI, §1 — Maximum principle

Theorem 11.1.1 (Maximum modulus theorem — interior form). *Conway VI.1.1; printed p. 128; Essential.*

If f is analytic in a region G and a is a point in G with $|f(a)| \geq |f(z)|$ for all z in G then f must be a constant function.

Used for. Minimum-modulus arguments, constant-range results, and FTA.

Theorem 11.1.2 (Maximum modulus theorem — boundary form). *Conway VI.1.2; printed p. 128; Essential.*

Let G be a bounded open set in $\mathbb{C}$ and suppose f is a continuous function on $\overline{G}$ which is analytic in G . Then

$$\max\{|f(z)| : z \in \overline{G}\} = \max\{|f(z)| : z \in \partial G\}$$

.

Used for. Boundary-modulus problems and reciprocal arguments.

Theorem 11.1.3 (Maximum modulus theorem — limsup form). *Conway VI.1.4; printed p. 129; Essential.*

Let G be a region in $\mathbb{C}$ and f an analytic function on G . Suppose there is a constant M such that $\limsup_{z \rightarrow a} |f(z)| \leq M$ for all a in $\partial_\infty G$. Then $|f(z)| \leq M$ for all z in G .

Used for. Boundary blow-up contradictions and Phragmén–Lindelöf exercises.

Exercise 11.1.4 (Minimum principle). *Conway VI.1, Exercise 1; printed p. 129; Supporting.*

Prove the following Minimum Principle. If f is a non-constant analytic function on a bounded open set G and is continuous on $\overline{G}$, then either f has a zero in G or $|f|$ assumes its minimum value on ∂G .

Used for. Interior minimum problems, bounded ratios, and reciprocal arguments.

11.2 Chapter VI, §2 — Schwarz's lemma

Theorem 11.2.1 (Schwarz lemma). *Conway VI.2.1; printed p. 130; Essential.*

Let $D = \{z : |z| < 1\}$ and suppose f is analytic on D with

- (a) $|f(z)| \leq 1$ for z in D ,
- (b) $f(0) = 0$. Then $|f'(0)| \leq 1$ and $|f(z)| \leq |z|$ for all z in the disk D . Moreover if $|f'(0)| = 1$ or if $|f(z)| = |z|$ for some $z \neq 0$ then there is a constant c , $|c| = 1$, such that $f(w) = cw$ for all w in D .

Used for. Disk estimates, half-plane fixed points, and equality cases.

Proposition 11.2.2 (Disk Möbius automorphisms). *Conway VI.2.2; printed p. 131; Important.*

If $|a| < 1$ then φ_a is a one-one map of $D = \{z : |z| < 1\}$ onto itself; the inverse of φ_a is φ_{-a} . Furthermore, φ_a maps ∂D onto ∂D , $\varphi_a(a) = 0$, $\varphi'_a(0) = 1 - |a|^2$, and $\varphi'_a(a) = (1 - |a|^2)^{-1}$, where $\varphi_a(z) = \frac{z - a}{1 - \bar{a}z}$.

Used for. Schwarz–Pick reductions and disk automorphism classification.

Theorem 11.2.3 (Classification of disk automorphisms). *Conway VI.2.5; printed p. 132; Important.*

Let $f : D \rightarrow D$ be a one-one analytic map of D onto itself and suppose $f(a) = 0$. Then there is a complex number c with $|c| = 1$ such that $f = c\varphi_a$.

Used for. Conformal self-maps and derivative bounds.

11.3 Chapter VI, §3 — Convex functions and Hadamard’s theorem

Theorem 11.3.1 (Three-lines theorem). *Conway VI.3.7; printed p. 135; Supporting.*

Let $a < b$ and let G be the vertical strip $\{x + iy : a < x < b\}$. Suppose $f : \overline{G} \rightarrow \mathbb{C}$ is continuous and f is analytic in G . If we define $M : [a, b] \rightarrow \mathbb{R}_+$ by $M(x) = \sup\{|f(x + iy)| : -\infty < y < \infty\}$, and $|f(z)| < B$ for all z in G , then $\log M(x)$ is a convex function.

Used for. Strip growth and logarithmic convexity.

Theorem 11.3.2 (Hadamard’s three-circles theorem). *Conway VI.3.13; printed p. 137; Important.*

Let $0 < R_1 < R_2 < \infty$ and suppose f is analytic and not identically zero on $\text{ann}(0; R_1, R_2)$. If $R_1 < r < R_2$ define $M(r) = \max\{|f(re^{i\theta})| : 0 \leq \theta \leq 2\pi\}$. Then for $R_1 < r_1 \leq r \leq r_2 < R_2$,

$$\log M(r) \leq \frac{\log r_2 - \log r}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log r - \log r_1}{\log r_2 - \log r_1} \log M(r_2)$$

.

Used for. Hadamard three-circles and Hardy-growth exercises.

11.4 Chapter VI, §4 — Phragmén–Lindelöf theorem

Theorem 11.4.1 (Phragmén–Lindelöf theorem). *Conway VI.4.1; printed p. 138; Important.*

Let G be a simply connected region and let f be an analytic function on G . Suppose there is an analytic function $\varphi : G \rightarrow \mathbb{C}$ which never vanishes and is bounded on G . If M is a constant and $\partial_\infty G = A \cup B$ such that:

- (a) for every a in A , $\limsup_{z \rightarrow a} |f(z)| \leq M$
- (b) for every b in B , and $\eta > 0$, $\limsup_{z \rightarrow b} |f(z)| |\varphi(z)|^\eta \leq M$; then $|f(z)| \leq M$ for all z in G .

Used for. Strip growth and exceptional-boundary-point problems.

Analytic convergence, normal families, and factorization

The convergence and construction theorems behind the advanced Part II problems.

12.1 Chapter VII, §1 — The space $C(G, \Omega)$

Definition 12.1.1 (Normal family). *Conway VII.1.14; printed p. 146; Definition.*

A set $\mathcal{F} \subset C(G, \Omega)$ is normal if each sequence in $\mathcal{F}$ has a subsequence which converges to a function f in $C(G, \Omega)$.

Used for. Understanding the conclusion of Arzelà–Ascoli and Montel.

Theorem 12.1.2 (Arzelà–Ascoli theorem). *Conway VII.1.23; printed p. 148; Important.*

A set $\mathcal{F} \subset C(G, \Omega)$ is normal iff the following two conditions are satisfied:

- (a) for each z in G , $\{f(z) : f \in \mathcal{F}\}$ has compact closure in Ω
- (b) $\mathcal{F}$ is equicontinuous at each point of G .

Used for. Normality in spaces of continuous functions.

12.2 Chapter VII, §2 — Spaces of analytic functions

Theorem 12.2.1 (Weierstrass convergence theorem). *Conway VII.2.1; printed p. 151; Essential.*

If $\{f_n\}$ is a sequence in $H(G)$ and f belongs to $C(G, \mathbb{C})$ such that $f_n \rightarrow f$ then f is analytic and $f_n^{(k)} \rightarrow f^{(k)}$ for each integer $k \geq 1$.

Used for. Closed-curve convergence, derivative families, and limits of analytic functions.

Corollary 12.2.2 (Completeness of $H(G)$). *Conway VII.2.3; printed p. 152; Supporting.*

$H(G)$ is a complete metric space.

Used for. Limits of Cauchy sequences of analytic functions.

Corollary 12.2.3 (Termwise differentiation in $H(G)$). *Conway VII.2.4; printed p. 152; Important.*

If $f_n : G \rightarrow \mathbb{C}$ is analytic and $\sum_{n=1}^{\infty} f_n(z)$ converges uniformly on compact sets to $f(z)$ then $f^{(k)}(z) = \sum_{n=1}^{\infty} f_n^{(k)}(z)$.

Used for. Infinite analytic series and function-space convergence.

Theorem 12.2.4 (Hurwitz's theorem). *Conway VII.2.5; printed p. 152; Essential.*

Let G be a region and suppose the sequence $\{f_n\}$ in $H(G)$ converges to f . If $f \not\equiv 0$, $\overline{B(a; R)} \subset G$, and $f(z) \neq 0$ for $|z - a| = R$ then there is an integer N such that for $n \geq N$, f and f_n have the same number of zeros in $B(a; R)$.

Used for. Moving preimages and limits of one-to-one functions.

Corollary 12.2.5 (Zero-free limits). *Conway VII.2.6; printed p. 152; Essential.*

If G is a region and $\{f_n\} \subset H(G)$ converges to f in $H(G)$ and each f_n never vanishes on G then either $f \equiv 0$ or f never vanishes.

Used for. Hurwitz applications and univalent-limit arguments.

Definition 12.2.6 (Locally bounded family). *Conway VII.2.7; printed p. 153; Definition.*

A set $\mathcal{F} \subset H(G)$ is locally bounded if for each point a in G there are constants M and $r > 0$ such that for all f in $\mathcal{F}$, $|f(z)| \leq M$ for $|z - a| < r$.

Used for. Applying Montel's theorem and proving compact-set bounds.

Lemma 12.2.7 (Local boundedness criterion). *Conway VII.2.8; printed p. 153; Important.*

A set $\mathcal{F}$ in $H(G)$ is locally bounded iff for each compact set $K \subset G$ there is a constant M such that $|f(z)| \leq M$ for all f in $\mathcal{F}$ and z in K .

Used for. Area-bound normality and composition of normal families.

Theorem 12.2.8 (Montel's theorem). *Conway VII.2.9; printed p. 153; Essential.*

A family $\mathcal{F}$ in $H(G)$ is normal iff $\mathcal{F}$ is locally bounded.

Used for. All normal-family questions in the qualifying exams.

12.3 Chapter VII, §4 — Riemann mapping theorem

Definition 12.3.1 (Conformal equivalence). *Conway VII.4.1; printed p. 160; Definition.*

A region G_1 is conformally equivalent to G_2 if there is an analytic function $f : G_1 \rightarrow \mathbb{C}$ such that f is one-one and $f(G_1) = G_2$.

Used for. Riemann mapping and classification of disk and half-plane maps.

Theorem 12.3.2 (Riemann mapping theorem). *Conway VII.4.2; printed p. 160; Essential.*

Let G be a simply connected region which is not the whole plane and let $a \in G$. Then there is a unique analytic function $f : G \rightarrow \mathbb{C}$ having the properties:

(a) $f(a) = 0$ and $f'(a) > 0$

(b) f is one-one

(c) $f(G) = \{z : |z| < 1\}$.

Used for. Disk/half-plane equivalence, symmetry, and punctured-domain questions.

12.4 Chapter VII, §5 — Weierstrass factorization

Lemma 12.4.1 (Uniform convergence of exponentials). *Conway VII.5.7; printed p. 166; Supporting.*

Let X be a set and let $f, f_1, f_2, \dots$ be functions from X into $\mathbb{C}$ such that $f_n(x) \rightarrow f(x)$ uniformly for x in X . If there is a constant a such that $\operatorname{Re} f(x) \leq a$ for all x in X then $\exp f_n(x) \rightarrow \exp f(x)$ uniformly for x in X .

Used for. Convergence of exponential factors.

Lemma 12.4.2 (Uniform convergence of infinite products). *Conway VII.5.8; printed p. 166; Supporting.*

Let (X, d) be a compact metric space and let $\{g_n\}$ be a sequence of continuous functions from X into $\mathbb{C}$ such that $\sum g_n(x)$ converges absolutely and uniformly for x in X . Then the product $f(x) = \prod_{n=1}^{\infty} (1 + g_n(x))$ converges absolutely and uniformly for x in X . Also there is an integer n_0 such that $f(x) = 0$ iff $g_n(x) = -1$ for some n , $1 \leq n \leq n_0$.

Used for. Functional equations and canonical products.

Theorem 12.4.3 (Analytic infinite-product theorem). *Conway VII.5.9; printed p. 167; Important.*

Let G be a region in $\mathbb{C}$ and let $\{f_n\}$ be a sequence in $H(G)$ such that no f_n is identically zero. If $\sum [f_n(z) - 1]$ converges absolutely and uniformly on compact subsets of G then $\prod_{n=1}^{\infty} f_n(z)$ converges in $H(G)$ to an analytic function $f(z)$. If a is a zero of f then a is a zero of only a finite number of the functions f_n , and the multiplicity of the zero of f at a is the sum of the multiplicities of the zeros of the functions f_n at a .

Used for. Infinite-product solutions and factorization of sine.

Definition 12.4.4 (Elementary factors). *Conway VII.5.10; printed p. 168; Definition.*

An elementary factor is one of the following functions $E_p(z)$ for $p = 0, 1, \dots$:

$$E_0(z) = 1 - z, \quad E_p(z) = (1 - z) \exp \left(z + \frac{z^2}{2} + \cdots + \frac{z^p}{p} \right), \quad p \geq 1.$$

Used for. Canonical products and Weierstrass factorization.

Theorem 12.4.5 (Canonical product theorem). *Conway VII.5.12; printed p. 169; Important.*

Let $\{a_n\}$ be a sequence in $\mathbb{C}$ such that $\lim |a_n| = \infty$ and $a_n \neq 0$ for all $n \geq 1$. (This is not a sequence of distinct points; but, by hypothesis, no point is repeated an infinite number of times.) If $\{p_n\}$ is any sequence of integers such that $\sum_{n=1}^{\infty} (r/|a_n|)^{p_n+1} < \infty$ for all $r > 0$ then $f(z) = \prod_{n=1}^{\infty} E_{p_n}(z/a_n)$ converges in $H(\mathbb{C})$. The function f is an entire function with zeros only at the points a_n . If z_0 occurs in the sequence $\{a_n\}$ exactly m times then f has a zero at $z = z_0$ of multiplicity m . Furthermore, if $p_n = n - 1$ then the preceding condition will be satisfied.

Used for. Entire functions with assigned discrete zero sets.

Theorem 12.4.6 (Weierstrass factorization theorem). *Conway VII.5.14; printed p. 170; Important.*

Let f be an entire function and let $\{a_n\}$ be the non-zero zeros of f repeated according to multiplicity; suppose f has a zero at $z = 0$ of order $m \geq 0$ (a zero of order $m = 0$ at $z = 0$ means $f(0) \neq 0$). Then there is an entire function g and a sequence of integers $\{p_n\}$ such that $f(z) = z^m e^{g(z)} \prod_{n=1}^{\infty} E_{p_n}(z/a_n)$.

Used for. Comparing functions with identical zeros and prescribed multiplicities.

Theorem 12.4.7 (Prescribed zeros on a region). *Conway VII.5.15; printed p. 170; Important.*

Let G be a region and let $\{a_j\}$ be a sequence of distinct points in G with no limit point in G ; and let $\{m_j\}$ be a sequence of integers. Then there is an analytic function f defined on G whose only zeros are at the points a_j ; furthermore, a_j is a zero of f of multiplicity m_j .

Used for. Natural-boundary constructions and all prescribed-zero exam problems.

Approximation, simple connectivity, and continuation

Runge, Mittag-Leffler, reflection, and monodromy.

13.1 Chapter VIII, §1 — Runge's theorem

Theorem 13.1.1 (Runge's theorem). *Conway VIII.1.7; printed p. 195; Essential.*

Let K be a compact subset of $\mathbb{C}$ and let E be a subset of $\mathbb{C}_\infty - K$ that meets each component of $\mathbb{C}_\infty - K$. If f is analytic in an open set containing K and $\varepsilon > 0$ then there is a rational function $R(z)$ whose only poles lie in E and such that $|f(z) - R(z)| < \varepsilon$ for all z in K .

Used for. Polynomial approximation, sign-function sequences, and approximation obstructions.

Corollary 13.1.2 (Runge approximation on open sets). *Conway VIII.1.14; printed p. 200; Important.*

Let G be an open subset of the plane and let E be a subset of $\mathbb{C}_\infty - G$ such that E meets every component of $\mathbb{C}_\infty - G$. Let $R(G, E)$ be the set of rational functions with poles in E and consider $R(G, E)$ as a subspace of $H(G)$. If $f \in H(G)$ then there is a sequence $\{R_n\}$ in $R(G, E)$ such that $f = \lim R_n$. That is, $R(G, E)$ is dense in $H(G)$.

Used for. Approximation on expanding compact sets with controlled poles.

Corollary 13.1.3 (Polynomial approximation on open sets). *Conway VIII.1.15; printed p. 200; Important.*

If G is an open subset of $\mathbb{C}$ such that $\mathbb{C}_\infty - G$ is connected then for each analytic function f on G there is a sequence of polynomials $\{p_n\}$ such that $f = \lim p_n$ in $H(G)$.

Used for. Approximation on expanding compact sets.

13.2 Chapter VIII, §2 — Simple connectedness

Theorem 13.2.1 (Equivalent forms of simple connectivity). *Conway VIII.2.2; printed p. 202; Essential.*

Let G be an open connected subset of $\mathbb{C}$. Then the following are equivalent:

- (a) G is simply connected
- (b) $n(\gamma; a) = 0$ for every closed rectifiable curve γ in G and every point a in $\mathbb{C} - G$
- (c) $\mathbb{C}_\infty - G$ is connected

- (d) for any f in $H(G)$ there is a sequence of polynomials that converges to f in $H(G)$
- (e) for any f in $H(G)$ and any closed rectifiable curve γ in G , $\int_{\gamma} f = 0$
- (f) every function f in $H(G)$ has a primitive
- (g) for any f in $H(G)$ such that $f(z) \neq 0$ for all z in G there is a function g in $H(G)$ such that $f(z) = \exp g(z)$
- (h) for any f in $H(G)$ such that $f(z) \neq 0$ for all z in G there is a function g in $H(G)$ such that $f(z) = [g(z)]^2$
- (i) G is homeomorphic to the unit disk
- (j) if $u : G \rightarrow \mathbb{R}$ is harmonic then there is a harmonic function $v : G \rightarrow \mathbb{R}$ such that $f = u + iv$ is analytic on G .

Used for. Branches on annuli, slit domains, primitives, roots, and Riemann mapping.

13.3 Chapter VIII, §3 — Mittag-Leffler's theorem

Theorem 13.3.1 (Mittag-Leffler theorem). *Conway VIII.3.2; printed p. 205; Essential.*

Let G be an open set, $\{a_k\}$ a sequence of distinct points in G without a limit point in G , and let $\{S_k(z)\}$ be the sequence of rational functions given by equation (3.1). Then there is a meromorphic function f on G whose poles are exactly the points $\{a_k\}$ and such that the singular part of f at a_k is $S_k(z)$.

Used for. Analytic Bézout identities and prescribed poles/zeros.

13.4 Chapter IX, §1 — Schwarz reflection principle

Theorem 13.4.1 (Schwarz reflection principle). *Conway IX.1.1; printed p. 210; Important.*

Let G be a region such that $G = G^*$. If $f : G_+ \cup G_0 \rightarrow \mathbb{C}$ is a continuous function which is analytic on G_+ and if $f(x)$ is real for x in G_0 , then there is an analytic function $g : G \rightarrow \mathbb{C}$ such that $g(z) = f(z)$ for z in $G_+ \cup G_0$.

Used for. Real/imaginary-axis symmetry and bounded half-plane problems.

13.5 Chapter IX, §2 — Analytic continuation along a path

Definition 13.5.1 (Function element and germ). *Conway IX.2.1; printed p. 214; Definition.*

A function element is a pair (f, G) where G is a region and f is an analytic function on G . For a given function element (f, G) define the germ of f at a to be the collection of all function elements (g, D) such that $a \in D$ and $f(z) = g(z)$ for all z in a neighborhood of a . Denote the germ by $[f]_a$.

Used for. Reading the Monodromy theorem and analytic-continuation statements.

13.6 Chapter IX, §3 — Monodromy theorem

Theorem 13.6.1 (Monodromy theorem). *Conway IX.3.6; printed p. 219; Important.*

Let (f, D) be a function element and let G be a region containing D such that (f, D) admits unrestricted continuation in G . Let $a \in D$, $b \in G$ and let γ_0 and γ_1 be paths in G from a to b ; let $\{(f_t, D_t) : 0 \leq t \leq 1\}$ and $\{(g_t, B_t) : 0 \leq t \leq 1\}$ be analytic continuations of (f, D) along γ_0 and γ_1 respectively. If γ_0 and γ_1 are FEP homotopic in G then $[f_1]_b = [g_1]_b$.

Used for. Path independence under fixed-end-point homotopy.

Corollary 13.6.2 (Global continuation on a simply connected region). *Conway IX.3.9; printed p. 220; Important.*

Let (f, D) be a function element which admits unrestricted continuation in the simply connected region G . Then there is an analytic function $F : G \rightarrow \mathbb{C}$ such that $F(z) = f(z)$ for all z in D .

Used for. Global branches and analytic continuation.

Harmonic and subharmonic functions

Mean values, maximum principles, Poisson integrals, Harnack convergence, and conformal invariance.

14.1 Chapter X, §1 — Basic properties

Definition 14.1.1 (Harmonic function). *Conway X.1.1; printed p. 251; Definition.*

If G is an open subset of $\mathbb{C}$ then a function $u : G \rightarrow \mathbb{R}$ is harmonic if u has continuous second partial derivatives and

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

This equation is called Laplace's equation.

Used for. Mean values, harmonic maximum principles, and Dirichlet problems.

Definition 14.1.2 (Harmonic conjugates). *Conway X.1.2; printed p. 251; Definition.*

If $f : G \rightarrow \mathbb{C}$ is an analytic function then $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$ are called harmonic conjugates.

Used for. Constructing analytic functions from harmonic real or imaginary parts.

Proposition 14.1.3 (Smoothness of harmonic functions). *Conway X.1.3; printed p. 252; Important.*

If $u : G \rightarrow \mathbb{R}$ is harmonic then u is infinitely differentiable.

Used for. Harmonic critical points, gradients, and conjugates.

Theorem 14.1.4 (Mean-value theorem). *Conway X.1.4; printed p. 253; Essential.*

If $u : G \rightarrow \mathbb{R}$ is a harmonic function and $\overline{B(a; r)}$ is a closed disk contained in G , then

$$u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{i\theta}) d\theta.$$

Used for. Harmonic Liouville, maximum principles, and area mean values.

Definition 14.1.5 (Mean Value Property). *Conway X.1.5; printed p. 253; Definition.*

A continuous function $u : G \rightarrow \mathbb{R}$ has the Mean Value Property (MVP) if whenever $\overline{B(a; r)} \subset G$,

$$u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{i\theta}) d\theta.$$

Used for. The harmonic maximum principles and converse mean-value theorem.

Theorem 14.1.6 (Maximum principle — first version). *Conway X.1.6; printed p. 253; Essential.*

Let G be a region and suppose that u is a continuous real valued function on G with the MVP. If there is a point a in G such that $u(a) \geq u(z)$ for all z in G then u is a constant function.

Used for. Interior extrema and harmonic open mapping.

Theorem 14.1.7 (Maximum principle — second version). *Conway X.1.7; printed p. 254; Essential.*

Let G be a region and let u and v be two bounded continuous real valued functions on G that have the MVP. If for each point a in the extended boundary $\partial_\infty G$, $\limsup_{z \rightarrow a} u(z) \leq \liminf_{z \rightarrow a} v(z)$, then either $u(z) < v(z)$ for all z in G or $u = v$.

Used for. Harmonic open mapping, uniqueness, and boundary-value problems.

Corollary 14.1.8 (Boundary uniqueness). *Conway X.1.9; printed p. 255; Important.*

Let G be a bounded region and suppose that $w : \overline{G} \rightarrow \mathbb{R}$ is a continuous function that satisfies the MVP on G . If $w(z) = 0$ for all z in ∂G then $w(z) = 0$ for all z in G .

Used for. Uniqueness for the Dirichlet problem.

Theorem 14.1.9 (Minimum principle). *Conway X.1.10; printed p. 255; Essential.*

Let G be a region and suppose that u is a continuous real valued function on G with the MVP. If there is a point a in G such that $u(a) \leq u(z)$ for all z in G then u is a constant function.

Used for. Nonnegative entire harmonic functions and reciprocal-style arguments.

Exercise 14.1.10 (Open mapping theorem for harmonic functions). *Conway X.1, Exercise 4; printed p. 255; Supporting.*

Prove that a nonconstant harmonic function on a region is an open map. (Hint: Use the fact that the connected subsets of $\mathbb{R}$ are intervals.)

Used for. August 2024 II.1.

Exercise 14.1.11 (Harmonicity of the logarithmic modulus). *Conway X.1, Exercise 5; printed p. 255; Supporting.*

If f is analytic on G and $f(z) \neq 0$ for any z show that $u = \log |f|$ is harmonic on G .

Used for. Boundary-modulus arguments, Jensen-type reasoning, and zero-free functions.

Exercise 14.1.12 (Area mean-value formula). *Conway X.1, Exercise 6; printed p. 255; Supporting.*

Let u be harmonic in G and suppose $\overline{B(a; R)} \subset G$. Show that

$$u(a) = \frac{1}{\pi R^2} \iint_{B(a; R)} u(x, y) \, dx \, dy$$

.

Used for. The L^1 -area-bound normal-family problem.

14.2 Chapter X, §2 — Harmonic functions on a disk

Theorem 14.2.1 (Poisson integral theorem). *Conway X.2.4; printed p. 257; Essential.*

Let $D = \{z : |z| < 1\}$ and suppose that $f : \partial D \rightarrow \mathbb{R}$ is a continuous function. Then there is a continuous function $u : \overline{D} \rightarrow \mathbb{R}$ such that

(a) $u(z) = f(z)$ for z in ∂D

(b) u is harmonic in D . Moreover u is unique and is defined by the formula

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta - t) f(e^{it}) dt$$

for $0 \leq r < 1$, $0 \leq \theta \leq 2\pi$.

Used for. Disk boundary-value and harmonic extension problems.

Corollary 14.2.2 (Poisson representation). *Conway X.2.9; printed p. 259; Important.*

If $u : \overline{D} \rightarrow \mathbb{R}$ is a continuous function that is harmonic in D then

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta - t) u(e^{it}) dt$$

for $0 \leq r < 1$ and all θ . Moreover, u is the real part of the analytic function

$$f(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{it} + z}{e^{it} - z} u(e^{it}) dt$$

.

Used for. Poisson formulas and harmonic conjugates.

Corollary 14.2.3 (Dirichlet problem on a disk). *Conway X.2.10; printed p. 259; Important.*

Let $a \in \mathbb{C}$, $p > 0$, and suppose h is a continuous real valued function on $\{z : |z - a| = p\}$; then there is a unique continuous function $w : \overline{B(a; p)} \rightarrow \mathbb{R}$ such that w is harmonic in $B(a; p)$ and $w(z) = h(z)$ for $|z - a| = p$.

Used for. Mean-value, boundary limits, and uniqueness.

Theorem 14.2.4 (Converse mean-value theorem). *Conway X.2.11; printed p. 260; Essential.*

If $u : G \rightarrow \mathbb{R}$ is a continuous function which has the MVP then u is harmonic.

Used for. Limits of harmonic functions and characterization by averages.

Theorem 14.2.5 (Harnack's inequality). *Conway X.2.14; printed p. 260; Essential.*

If $u : \overline{B(a; R)} \rightarrow \mathbb{R}$ is continuous, harmonic in $B(a; R)$, and $u \geq 0$ then for $0 \leq r < R$ and all θ ,

$$\frac{R-r}{R+r}u(a) \leq u(a + re^{i\theta}) \leq \frac{R+r}{R-r}u(a)$$

.

Used for. Quantitative comparison for positive harmonic functions.

Theorem 14.2.6 (Harnack's theorem). *Conway X.2.16; printed p. 261; Essential.*

Let G be a region.

(a) The metric space $\text{Har}(G)$ is complete.

(b) If $\{u_n\}$ is a sequence in $\text{Har}(G)$ such that $u_1 \leq u_2 \leq \dots$ then either $u_n(z) \rightarrow \infty$ uniformly on compact subsets of G or $\{u_n\}$ converges in $\text{Har}(G)$ to a harmonic function.

Used for. Compactness and convergence of harmonic functions.

14.3 Chapter X, §3 — Subharmonic functions

Definition 14.3.1 (Subharmonic and superharmonic functions). *Conway X.3.1; printed p. 264; Definition.*

Let G be a region and let $\varphi : G \rightarrow \mathbb{R}$ be a continuous function. φ is a subharmonic function if whenever $\overline{B(a; r)} \subset G$,

$$\varphi(a) \leq \frac{1}{2\pi} \int_0^{2\pi} \varphi(a + re^{i\theta}) d\theta.$$

φ is a superharmonic function if whenever $\overline{B(a; r)} \subset G$,

$$\varphi(a) \geq \frac{1}{2\pi} \int_0^{2\pi} \varphi(a + re^{i\theta}) d\theta.$$

Used for. Subharmonic maximum principles, comparison, and Perron arguments.

Theorem 14.3.2 (Maximum principle — third version). *Conway X.3.2; printed p. 264; Essential.*

Let G be a region and let $\varphi : G \rightarrow \mathbb{R}$ be a subharmonic function. If there is a point a in G with $\varphi(a) \geq \varphi(z)$ for all z in G then φ is a constant function.

Used for. Maximum arguments for subharmonic functions.

Theorem 14.3.3 (Maximum principle — fourth version). *Conway X.3.3; printed p. 264; Essential.*

Let G be a region and let φ and ψ be bounded real valued functions defined on G such that φ is subharmonic and ψ is superharmonic. If for each point a in $\partial_\infty G$, $\limsup_{z \rightarrow a} \varphi(z) \leq \liminf_{z \rightarrow a} \psi(z)$, then either $\varphi(z) < \psi(z)$ for all z in G or $\varphi = \psi$ and φ is harmonic.

Used for. Boundary comparison for subharmonic functions.

Theorem 14.3.4 (Harmonic comparison characterization). *Conway X.3.4; printed p. 265; Important.*

Let G be a region and $\varphi : G \rightarrow \mathbb{R}$ a continuous function. Then φ is subharmonic iff for every region G_1 contained in G and every harmonic function u_1 on G_1 , $\varphi - u_1$ satisfies the Maximum Principle on G_1 .

Used for. Recognizing subharmonic functions by harmonic comparison.

Corollary 14.3.5 (Boundary comparison characterization). *Conway X.3.5; printed p. 265; Important.*

Let G be a region and $\varphi : G \rightarrow \mathbb{R}$ a continuous function; then φ is subharmonic iff for every bounded region G_1 such that $\overline{G_1} \subset G$ and for every continuous function $u_1 : \overline{G_1} \rightarrow \mathbb{R}$ that is harmonic in G_1 and satisfies $\varphi(z) \leq u_1(z)$ for z on ∂G_1 , $\varphi(z) \leq u_1(z)$ for z in G_1 .

Used for. Subharmonic pullbacks and Perron-family arguments.

Corollary 14.3.6 (Maximum of subharmonic functions). *Conway X.3.6; printed p. 265; Supporting.*

Let G be a region and φ_1 and φ_2 subharmonic functions on G ; if $\varphi(z) = \max\{\varphi_1(z), \varphi_2(z)\}$ for each z in G then φ is a subharmonic function.

Used for. Patching subharmonic barriers.

Corollary 14.3.7 (Harmonic replacement). *Conway X.3.7; printed p. 266; Supporting.*

Let φ be a subharmonic function on a region G and let $\overline{B(a; r)} \subset G$. Let φ' be the function defined on G by:

$$(i) \quad \varphi'(z) = \varphi(z) \text{ if } z \in G - B(a; r)$$

$$(ii) \quad \varphi' \text{ is the continuous function on } \overline{B(a; r)} \text{ which is harmonic in } B(a; r) \text{ and agrees with } \varphi(z) \text{ for } |z - a| = r. \text{ Then } \varphi' \text{ is subharmonic.}$$

Used for. Patching subharmonic functions with harmonic replacements.

Exercise 14.3.8 (Conformal invariance of subharmonicity). *Conway X.3, Exercise 6; printed p. 267; Supporting.*

If $f : G \rightarrow \Omega$ is analytic and $\varphi : \Omega \rightarrow \mathbb{R}$ is subharmonic, show that $\varphi \circ f$ is subharmonic if f is one-one. What happens if $f'(z) \neq 0$ for all z in G ?

Used for. January 2025 II.1.

Range theorems for entire functions

The one later result needed by a recorded qualifying-exam problem.

15.1 Chapter XII, §2 — Little Picard theorem

Theorem 15.1.1 (Little Picard theorem). *Conway XII.2.3; printed p. 296; Essential.*

If f is an entire function that omits two values then f is a constant.

Used for. Surjectivity of $(f^3 + f^2)(\mathbb{C})$ for nonconstant entire f .

Part III

Indexes

Named Results

The named results are divided into personal and non-personal names. Each entry links to its full statement in the reference.

16.1 Results Named after a Person

- **II.3.7.** Cantor intersection theorem
- **II.4.8.** Lebesgue covering lemma
- **II.4.10.** Heine–Borel theorem
- **II.5.15.** Heine–Cantor theorem
- **II.6.2.** Weierstrass M-test
- **III.2.24.** Cauchy–Riemann equations
- **III.2.24** — **note.** Differentiability implies Cauchy–Riemann
- **III.2.29.** Cauchy–Riemann criterion
- **III.3.14.** Möbius maps preserve generalized circles
- **IV.2.13.** Cauchy formula for derivatives
- **IV.2.14.** Cauchy estimates
- **IV.3.4.** Liouville’s theorem
- **IV.5.4.** Cauchy’s integral formula — first version
- **IV.5.6.** Cauchy’s integral formula — second version
- **IV.5.7.** Cauchy’s theorem — first version
- **IV.5.8.** Cauchy formula for derivatives — several curves
- **IV.5.9.** Cauchy formula for derivatives — one curve
- **IV.5.10.** Morera’s theorem
- **IV.6.6.** Cauchy’s theorem — second version
- **IV.6.7.** Cauchy’s theorem — third version
- **IV.6.15.** Cauchy’s theorem — fourth version
- **IV.§8.** Goursat’s theorem
- **V.1.21.** Casorati–Weierstrass theorem
- **V.3.8.** Rouché’s theorem
- **V.3.8a.** Classical Rouché inequality
- **VI.2.1.** Schwarz lemma
- **VI.3.13.** Hadamard’s three-circles theorem
- **VI.4.1.** Phragmén–Lindelöf theorem
- **VII.1.23.** Arzelà–Ascoli theorem
- **VII.2.1.** Weierstrass convergence theorem
- **VII.2.5.** Hurwitz’s theorem
- **VII.2.9.** Montel’s theorem
- **VII.4.2.** Riemann mapping theorem
- **VII.5.14.** Weierstrass factorization theorem
- **VIII.1.7.** Runge’s theorem
- **VIII.1.14.** Runge approximation on open sets
- **VIII.3.2.** Mittag-Leffler theorem
- **IX.1.1.** Schwarz reflection principle
- **X.2.4.** Poisson integral theorem

- **X.2.14.** Harnack's inequality
- **X.2.16.** Harnack's theorem
- **XII.2.3.** Little Picard theorem

16.2 Other Named Results

- **II.5.10.** Intermediate-value property
- **II.5.12.** Extreme-value theorem
- **II.6.1.** Uniform limit theorem
- **IV.1.18.** Fundamental theorem for complex line integrals
- **IV.3.5.** Fundamental theorem of algebra
- **IV.3.7.** Identity theorem
- **IV.3.8.** Identity theorem for two functions
- **IV.7.5.** Open mapping theorem
- **IV.7.6.** Global analytic inverse theorem
- **V.2.2.** Residue theorem
- **V.3.4.** Argument principle
- **VI.1.1.** Maximum modulus theorem — interior form
- **VI.1.2.** Maximum modulus theorem — boundary form
- **VI.1.4.** Maximum modulus theorem — limsup form
- **VI.1, Exercise 1.** Minimum principle
- **VI.3.7.** Three-lines theorem
- **IX.3.6.** Monodromy theorem
- **X.1.4.** Mean-value theorem
- **X.1.6.** Maximum principle — first version
- **X.1.7.** Maximum principle — second version
- **X.1.10.** Minimum principle
- **X.2.11.** Converse mean-value theorem
- **X.3.2.** Maximum principle — third version
- **X.3.3.** Maximum principle — fourth version

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